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Blood flow distribution in an anatomically detailed arterial network model: criteria and algorithms

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Abstract Development of blood flow distribution criteria is a mandatory step towards developing computational models and numerical simulations of the systemic circulation. In the present work we (i) present a systematic approach based on anatomical and physiological considerations to distribute the blood flow in a one-dimensional anatomically detailed model of the arterial network, and (ii) develop a numerical procedure to calibrate resistive parameters in terminal models in order to effectively satisfy such flow distribution. For the first goal, we merge data collected from the specialized medical literature with anatomical concepts such as vascular territories to determine blood flow supply to specific (encephalon, kidneys, etc.) and distributed (muscles, skin, etc.) organs. Overall, 28 entities representing the main specific organs are accounted for in the detailed description of the arterial topology that we use as model substrate. In turn, 116 vascular territories are considered as the basic blocks that compose the distributed organs throughout the whole body. For the second goal, Windkessel models are used to represent the peripheral beds, and the values of the resistive parameters are computed applying a Newton method to a parameter identification problem to guarantee the supply of the correct flow fraction to each terminal location according to the given criteria. Finally, it is shown that, by means of the criteria developed, and for a rather standard set of model parameters, the model predicts physiologically realistic pressure and flow waveforms.

Keywords Computational hemodynamics · Anatomy · Vascular territories · Arterial circulation · Parameter identification

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1 Introduction

Through the decades, simulation of blood flow has been accomplished using different levels of model sophistication, from lumped to distributed, according to the level of detail required by applications. In view of the complex propagatory phenomena that yield the characteristic blood pressure and flow records, it has been widely acknowledged that an adequate approach to predict accurate hemodynamic environments demands a global conception of the systemic circulation. Under this premise, a simplified approach based on the use of one-dimensional (1D) models remains the best choice in trade-off between computational cost and predictive modeling capabilities. These simple models have been

successfully used in understanding and predicting the impact of abnormal conditions on the systemic response such as arterial stenoses, bypass grafts and hypertension conditions, among others (Reymond et al, 2012; Steele et al, 2003; Stergiopulos et al, 1992; Van de Vosse and Stergiopulos, 2011; Vardoulis et al, 2011). Such models have also been used in elucidating the characteristics of arterial transfer functions to assess central pressure from peripheral measurements (Stergiopulos et al, 1998), to evaluate variability in hemodynamics due to anatomical variations (Alastruey et al, 2008a), and to study how to relate arterial stiffenning and pulse wave velocity using non-invasive measurements (Gaddum et al, 2013). Moreover, given that it has been largely recognized that the three-dimensionality of blood flow is responsible for (or at least closely related to) the onset and progress of several cardiovascular diseases (Caro, 2009; Caro et al, 1996, 1969, 1971; Coskun et al, 2006; Friedman et al, 1986; Fry, 1968; Kornet et al, 1999), simplified, but accurate, 1D models are important in providing consistent boundary conditions to set 3D blood flow simulations (Blanco et al, 2009, 2010, 2012a; Kim et al, 2009, 2010; Migliavacca et al, 2006).

To date, we can mention the works of Anliker et al (1971a,b); Noordergraaf et al (1963); Schaaf and Abbrecht (1972); Spencer and Deninson (1959); Westerhof et al (1969); Avolio (1980); Sherwin et al (2003); Stergiopulos et al (1992); Stettler et al (1981a,b); Alastruey et al (2011); Matthys et al (2007); Olufsen et al (2000); Reymond et al (2009) in which reasonably detailed descriptions of the arterial network are provided. However, even in such descriptions, the networks are ultimately truncated at a certain -high- level of arterial branching, also truncating the validity of the predicted hemodynamics at smaller vessels given the strong constrains regarding the pressure-flow relation which are imposed by lumped models at terminal locations. This has motivated the development of an anatomically detailed arterial network model, as proposed in this work.

Blood flow simulations in an anatomically detailed arterial network can shed light on developing new cardiovascular indexes towards establishing robust and efficient characterization of cardiovascular diseases, as well as to guide therapeutic actions. Current indexes derived from pressure waveforms, or from propagation phenomena are not enough to evaluate arterial function because the interplay among structural properties of arterial wall, morphometry, anatomy and, ultimately, arterial diseases continues to be poorly understood (Westerhof and Westerhof, 2013).

Also, applications in which the proposed highly detailed model can bring solutions is on surgical procedures, such as bypass grafts. These actions result in changes in the arterial topology, sometimes altering significantly blood perfusion, and waveforms, which may lead to deviations from control conditions, provoking the onset of abnormal hemodynamic conditions. For example, in certain patients, radial harvesting procedures for coronary revascularization can lead to neurological complications and tissue necrosis (Denton et al, 2001; Lee et al, 2004), stressing the need for an accurate hemodynamics assessment to guarantee adequate blood supply to the hand. Analogously, proper hemodynamic evaluation of conditions such as those observed in aortic coarctation, or aorto-iliac disease requires considering collateral pathways (intercostal arteries to aorta artery, internal thoracic to inferior epigastric artery, etc.) which start to play a major role in the perfusion of the lower limb (Yurdakul et al, 2006).

A more critical example in which the present model will contribute is in the evaluation of the spinal cord hemodynamics, whose sensitivity to flow disruption is very high (Duggal and Lach, 2002). The blood supply to the spine is performed through a highly collateralized network of small arteries which cannot be easily accessed to obtain measurements. Therefore, there is very little known about the charcteristics of the spinal cord hemodynamics (see for example Martin et al (2012)). For instance, it was shown that some cardiovascular events such as cardiac arrest, or occlusion of radicular arteries at insufficiently collateralized locations strongly affect the oxygenation of the spinal cord, leading to neuronal loss and even spinal cord infarction (Novy et al, 2006). Therefore, it is of the utmost importance to understand in detail the blood flow supply to the spinal cord. In this context, results obtained from anatomically detailed blood flow simulations would help to provide a more realistic hemodynamics assessment for, among others, the prediction of perivascular flow in the spinal cord, with strong implications in the study of craniospinal disorders (Bilston et al, 2010; Elliott et al, 2011).

One of the main problems in setting such computational models of the arterial hemodynamics is the definition of the blood flow distribution, or equivalently, the setting of outflow boundary conditions at terminal locations. In all the works cited above, this task is somehow accomplished resorting to physiological criteria. Nevertheless, due to the truncation of the network, no direct, clear and complete relation between detailed blood flow delivery and structural conformation of the anatomical entities (organs, territories, etc.) is present.

To determine outflow boundary conditions, in Schaaf and Abbrecht (1972); Stergiopulos et al (1992); Stettler et al (1981a) terminal resistances are used to meet a

given blood flow distribution. An alternative strategy is to employ bifurcation laws (Murray, 1926; Uylings, 1977) to relate flow distribution with arterial radii. For example, in Olufsen (1999) a semi-analytical recursive method is used to construct such peripheral impedances, while Schreiner and Buxbaum (1993) exploit these laws to generate vascular trees. In Grinberg and Karniadakis (2008), outflow conditions are considered using time-dependent resistances, but this is doable in view of the truncation of the arterial tree at first or second generation of arteries. Similarly, in Alastruey et al (2008a) (as well as in many other contributions involving regulation mechanisms) time-dependent resistances are used to regulate blood flow in the brain. Finally, in Troianowski et al (2011), an iterative procedure based on a fixed point method is used to enforce a given blood flow distribution in patients with cavopulmonary connection.

In this work we address the problem of determining flow supply within a highly detailed arterial network which incorporates almost all the arteries described by well-known anatomy textbooks (see Dauber (2007); Netter (2011); Watanabe (2013) and Watanabe et al (2013) for a preliminary stage of the model). As a first characterization of our approach, destination of blood is split into two departments: *specific* organs, such as the kidneys, the encephalon, the liver, the eyes, the bladder, and every organ with a bounded localization within the body, and *distributed* organs, such as skin, muscles and bones, which are all along the body. Data of blood requirements to specific organs can be relatively easy to find in the specialized literature. However, data concerning blood supply to distributed organs is reported without providing differentiated regional distribution. This is limiting when setting regional blood flow supply in a highly detailed arterial network such as the one employed here.

The aim of the present work is twofold: we (i) present a systematic approach based on stringent anatomical and physiological considerations to distribute the blood flow in our one-dimensional anatomically detailed model of the arterial network called ADAN (which stands for *Anatomically Detailed Arterial Network* see Watanabe (2013)), and (ii) develop a numerical procedure to calibrate terminal resistive parameters present in the ADAN model, such that its hemodynamic response effectively follows a given flow distribution.

Regarding the first goal, we present the necessary criteria to determine blood flow distribution in model of the arterial network with over 1500 arterial vessels. These criteria are: retrieved from the literature in the case of specific organs whenever possible, and developed from merging anatomical data and physiological hypotheses in the case of distributed organs. In the distributed case, this yields a systematic approach that is supported by the concept of vascular territories and segmental laws Pavol et al (2002). Such laws are constructed to provide relations between the masses of the different segments of the body (head, arm, hand, foot, trunk, etc.). Vascular territories are employed to understand the body as a 3D jigsaw puzzle concerning such distributed entities. For each of these unit blocks, a correspondence with the feeding vessel, the so-called source artery, is established, and fractions of area are determined. These area fractions are converted into tissue volume fractions through the segmental laws. Finally, a power law relation between volume fraction and flow fraction is hypothesized. The power law hypothesis is crucial towards bridging the microvascular features with the blood flow supply. This is very well supported by experimental and analytical studies (Francis et al, 2009; Cormack and Lamberty, 1986; West et al, 1997).

Concerning the second goal, a numerical strategy is proposed and implemented to calibrate model parameters at terminal locations (the resistive elements in Windkessel models) to guarantee that the proposed blood flow distribution criteria are satisfied within the context of 1D hemodynamic simulations. Flow distribution is understood in the following sense: in the periodic state, the average flow fraction in a cardiac cycle is as specified by the theoretical criteria. This is not a trivial task due to the viscous effects introduced by the upstream arterial vasculature. This numerical approach relies on writing the problem of identification of peripheral vascular resistances as a system of non-linear equations. This approach was presented in Blanco et al (2012b) to compute peripheral resistances in a detailed network of the vasculature of the arm using the Broyden method. In the present case, the complexity of the model and the amount of unknowns requires using a Newton method and a fully parallel computational implementation, without which the identification of resistances becomes unfeasible.

We show that with this approach the model predicts physiologically realistic pressure and flow waveforms, and puts in evidence the need for an iterative method to compute effectively the terminal model parameters in a realistic case with viscous blood flow for which a simple analytical solution to determine terminal resistances gives large deviations in the flow distribution resulting from simulations with respect to the distribution defined through the theoretical criteria.

The organization of this work is the following. In Section 2 the features of the arterial network used as model substrate are outlined. An account of the specific organs considered in the model is given in Section 3, while the corresponding description of distributed or-

gans is in Section 4. The numerical approach to guarantee such blood distribution is described in Section 5, and the numerical results are described in Section 6. The final remarks are developed in Section 7.

2 Arterial model overview

2.1 Network topology

Let us give a brief description of the arterial network used as model substrate on top of which the blood flow distribution guidelines will be applied. An assessment of the anatomical fidelity of the present model can be found in Watanabe (2013).

In the topological definition of the arterial network we consider 2142 arteries (1598 arterial segments with a well-acknowledged name and 544 perforator arteries), which were drawn in 3D space using a digital dataset of the human skeleton as scaffold. Such arterial network is fully compliant with descriptive anatomy textbooks (Dauber, 2007; Netter, 2011) for the case of an average young healthy male. The diameter of each arterial vessel was obtained from anatomical/medical literature. This constitutes the basis for the estimation of model parameters. Parameters describing material properties of arterial vessels are estimated using data reported in the literature. We remark that this is not primary in the present work.

The proposed model incorporates the entire vasculature up to the pre-arteriolar level. At such level the so-called perforator arteries are included. The number of specific organs that receive blood is 28, while the number of vascular territories (to be introduced in Section 4) is 116. Thus, overall the blood is delivered to $28 + 116 = 144$ peripheral beds, which are supplied by one or more delivery and/or perforator arteries.

In Table 1 the global features of the arterial network and a relation with global human body features is presented, and Figure 1 presents several snapshots of the complex arterial topology used as underlying model for the developments of the present contribution.

Height [cm]	170
Body surface area [m^2]	1.6488
Weight* [kg]	56
Arterial segments	1598
Number of specific organs	28
Number of vascular territories	116
Number of perforator arteries	544

Table 1 General features of the arterial network and the relation with human body characteristics. * According to the formula BSA = $0.007184\mathrm{W}^{0.425}\mathrm{H}^{0.725}$ (Verbraecken et al, 2006).

2.2 Governing equations

We employ the classical 1D governing equations for flow in compliant pipes describing flow rate Q, lumen area A and blood pressure P (Hughes, 1973)

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0, \tag{1}$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right) = -\frac{A}{\rho}\frac{\partial P}{\partial x} + \frac{8\pi\mu}{\rho A}Q, \tag{2}$$

where ρ and μ are blood density and viscosity, respectively. The constitutive arterial wall equation accounts for elastin, collagen and smooth muscle contributions (Armentano et al, 1991; Urquiza et al, 1995)

$$P = P_o + \frac{\pi h R_o}{A}\left[E_E\varepsilon + E_C\epsilon_r \ln(\mathrm{e}^u + 1) + K_M\dot{\varepsilon}\right], \tag{3}$$

with $\varepsilon = \sqrt{\frac{A}{A_o}} - 1$ and $u = \frac{\varepsilon - \varepsilon_0}{\epsilon_r}$. Here, h is the wall thickness E_E is the effective elastic modulus of elastin, E_C the effective elastic modulus of collagen (ϵ_r and ε_0 characterize the recruitment of fibers) and K the effective viscoelastic parameter.

Regarding viscous effects we have considered the Poiseuille flow in the entire tree. Evidently, this is a poor model when compared with more sophisticated strategies such as in Reymond et al (2009), but even in such situation the model was able to deliver realistic results (see Section 6). The mass conservation equation was written without mass leakage. Existing models have used the leaking artery approach to model the effect of blood leaving the aorta towards the intercostal arteries. In our model, the entire intercostal circuits are present. So, there is no need to regard leakage in the aorta. The same happens with the peripheral vasculature. The existence of perforator arteries provide a rather uniform blood supply from source arteries to vascular territories.

At bifurcations the coupling equations are

$$\sum_{i=1}^{N_T} Q_i = 0, \tag{4}$$

$$P_i = P_1 \quad 2, \ldots, N_T. \tag{5}$$

with N_T the number of converging arterial segments.

At terminal arterial segments, Windkessel elements are considered to incorporate the effect of the remaining vasculature.

$$R_A R_B C\frac{dQ}{dt} = R_B C\frac{d}{dt}(P - P_T) + (P - P_T) - (R_A + R_B)Q, \tag{6}$$

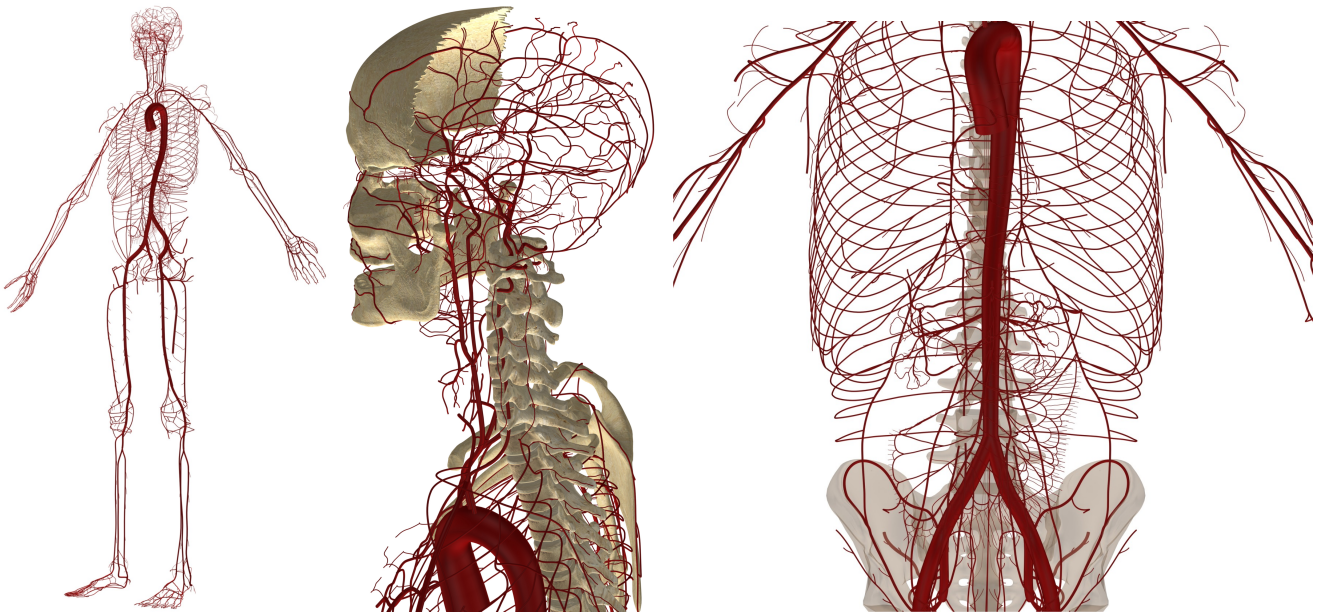


Fig. 1 Arterial topology on top of which refined blood flow distribution criteria are developed. For more details see Watanabe (2013).

where P_T is a reference terminal pressure, C is the peripheral compliance and R_A and R_B stand for the resistive elements. The resistance of a peripheral bed is $R_T = R_A + R_B$, and we assume $R_A = \theta_R R_T$, with $\theta_R = 0.2$ (Stergiopulos et al, 1992). Territory specific relations between R_A and R_T are being matter of research by exploring vascular arrengements obtained through the use of the so-called *constrained constructive optimization* algorithms (Blanco et al, 2013; Karch et al, 1999).

The set of parameters present in equation (3), as well as the peripheral compliance and terminal pressure $\{C, P_T\}$, impact on the response of the computational model. Nevertheless, the proposed blood delivery criteria and the numerical strategy are independent from these parameters. In other words, having fixed all parameters, the present contribution is directed towards defining the resistive parameters R_T at each terminal location of the 1D model shown in Figure 1.

2.3 Model parameters

The setting of model parameters has been part of the present research. As stated before, the focus of this paper is the methodology to estimate terminal parameters such a given blood flow distribution is met. Model parameter determination/justification and verifications are the matter of a forthcoming paper. However, in this section we limit ourselves to provide the basic data that was used in the simulations.

The complete model dataset has been made publicly available at **http://hemolab.lncc.br/adan-web**. Arterial radii were extracted from the literature. Wall thickness was estimated based on curve-fitting of data published by Avolio (1980), leading to

$$\frac{h}{R_o} = a\mathrm{e}^{(bR_o)} + c\mathrm{e}^{(dR_o)}, \tag{7}$$

with $a = 0.2802$, $b = -5.053$ 1/cm, $c = 0.1324$ and $d = -0.1114$ 1/cm.

Material parameters have been established following a rule of mixtures approach. The corresponding proportions W_E, W_C, W_M are such that $W_E + W_C + W_M = 1$ and then we have $E_E = W_E E_e$, $E_C = W_C E_c$ and $K_M = W_M K_m$. The parameters have been taken from Avolio (1980); Weizsacker and Pinto (1988); Armentano et al (1991), and are summarized in Table 2.

Parameter	Value
E_e [dyn/cm^2]	$4 \cdot 10^6$
E_c [dyn/cm^2]	$1 \cdot 10^9$
ε_0	0.35
ϵ_r	0.05
K_m [dyn s/cm^2]	$3 \cdot 10^5$

Table 2 Values of material properties for the arterial wall.

Based on data published in (Burton, 1954; Slutskii and Purinya, 1974), arterial vessels were classified into groups according to the vessel radius, as shown in Table 3, and a fraction of tissue was determined as given in Table 4.

Group	Lumen radius range, R_o [cm]
A	$R_o > 0.18$
B	$0.07 \leq R_o \leq 0.18$
C	$0.07 \geq R_o$

Table 3 Arterial vessel groups according to the lumen radius.

Group	A	B	C
W_E	0.9	0.7	0.45
W_C	0.099	0.25	0
W_M	0.001	0.05	0.55

Table 4 Fractions of wall constituents in each of the three groups considered.

Blood viscosity was taken $\mu = 4$ cP for arteries and $\mu = 1$ cP for perforators (emulating plasma viscosity) in order to avoid exaggerated and unrealistic pressure drop in the smallest vessels. Blood density was considered to be $\rho = 1.04$ g/cm^3.

Finally, peripheral compliance was taken to be 10% of arterial tree compliance (computed just with the linear elastic contribution in (3)). This value was considered to be a smaller fraction than that used in Stergiopulos et al (1992) (there, it was 16.5%), due to having more arterial segments in our model.

3 Blood supply to specific organs

We define specific organs as the blood destinations which are located in a bounded region of the human body. In this classification we basically have all internal organs such as eyes, heart, kidneys, gallbladder and so on (see Table 5). These organs receive blood from one or more arterial vessels, and although the concept of vascular territories (see Section 4) also applies to intraorgan blood distribution, we do not exploit this concept at the specific organ level in the present work.

In Table 5 the list of specific organs considered in the realization of the arterial network for the case of an average young healthy male is given. The blood flow fraction (BFF) to each organ is obtained from the corresponding reference (or group of references) cited in such list. In these works, data is reported either in direct percentage form (percentage data, PD), as relative blood flow per 100g of mass of tissue (mass data, MD), which must be complemented with the mass of the tissue perfused, or in terms of blood velocity (velocity data, VD). In the case of the diaphragm, the lack of data led us to consider the Murray law (law data, LD) for the estimation of its blood supply. The denomination given in Table 5 will be used for ease of reference in the presentation of the results.

In the present model, organs are supplied by one or more arterial segments as given by the value of NP in Table 5. Thus, an intra-organ distribution (IOD) criterion is in order. Also in such table, IOD is detailed for each specific organ. Two approaches have been pursued in the case of more than one arterial vessel supplying blood: in the case of blood delivery through perforator arteries (see this concept in Section 4) an equal distribution is established for each perforator (code IE); in the case of blood supplied to the encephalon, kidneys and the heart, the distribution is set based on the Murray law (code IB). These criteria can be further refined provided more anatomical data is added to the model regarding the microcirculatory characteristics within the different organ tissues.

In total, the specific organs require approximately 64.7% of the cardiac output. The complementary flow is supplied to the distributed organs as detailed in the following section.

4 Blood delivery to vascular territories

From Table 5 it turns out that there is, approximately, a remaining 35.3% of blood flow to be specified. Such percentage is supplied to the distributed organs, namely: the muscles, bones, nerves, fascia and skin. This number, which is derived from the data compiled in Table 5 is consistent with the data reported in Valentin (2002); Mohrman and Heller (2006). The problem addressed in this section is the differentiated blood supply to these structures. This is achieved by considering a unique distributed organ which embodies all these anatomical structures. For this distributed organ we exploit the concept of vascular territories.

4.1 Historical review

Let us start reviewing the research efforts that led to the establishment of the concept of vascular territories and subsequently let us provide the guidelines to define the regional blood supply.

Firstly, Carl Manchot used dissection to identify the main arterial vessels supplying 40 pieces which composed a first vascular map of the surface of part the human skin (see Manchot (2011)). Posteriorly, the X-ray technology allowed Michel Salmon to perform injection studies followed by tomography to improve the accuracy in the characterization of the path of cutaneous vessels, separating the skin surface into 80 pieces (see Salmon et al (1988)). However, it was just in Taylor and Palmer (1987) that the conceptual bases of modern peripheral vascular anatomy were established, addressing the characterization of blood supply not only to the skin, but also to deep structures. In the aforementioned

Code	Organ	BFF [%]	DT	Work	NP	IOD
O1	Heart	4	PD	(Valentin, 2002)	18	IB
O2	Encephalon	12	PD	(Valentin, 2002)	130	IB
O3/O4	Eye (×2)	0.014286	PD	(Williamson and Harris, 1994)	4	IE
O5/O6	Ear (×2)	0.000014	PD	(Vance and Prazma, 1984)	4	IE
O7	Nose	0.000089	MD	(Bende and Flisberg, 1985; Valentin, 2002)	10	IE
O8	Tongue	0.3	MD	(Bergman et al, 1982; Valentin, 2002)	8	IE
O9	Teeth	0.0012	VD	(Yoon et al, 2010)	32	IE
O10	Thyroid	1.5	PD	(Valentin, 2002)	10	IE
O11	Hypophysis	0.009429	MD	(McIntosh and Anderson, 2010; Valentin, 2002)	4	IE
O12	Liver	6.5	PD	(Valentin, 2002)	8	IE
O13	Gallbladder	0.004286	MD	(McIntosh and Anderson, 2010; Valentin, 2002)	1	IE
O14/O15	Kidney (×2)	9.5	PD	(Valentin, 2002)	49	IB
O16/17	Suprarenal (×2)	0.15	PD	(Valentin, 2002)	3	IE
O18	Stomach	1	PD	(Valentin, 2002)	17	IE
O19	Pancreas	1	PD	(Valentin, 2002)	6	IE
O20	Spleen	3	PD	(Valentin, 2002)	1	IE
O21	Small intestine	10	PD	(Valentin, 2002)	64	IE
O22	Large intestine	3.25	PD	(Valentin, 2002)	86	IE
O23	Bladder	0.06	PD	(Valentin, 2002)	2	IE
O24	Penis	0.893140	VD	(Halls et al, 2009)	2	IE
O25/O26	Testicle (×2)	0.028750	MD	(McIntosh and Anderson, 2010; Valentin, 2002)	1	IE
O27	Rectum	0.75	PD	(Valentin, 2002)	5	IE
O28	Diaphragm	1.058718	LD	(Loukas et al, 2005)	2	IE
	Total	64.712962				

Table 5 Description of specific organs and their blood supply. BFF: blood flow fraction in %, DT: data type, namely: PD: data is directly reported in percentage, MD: data is reported in blood flow per 100g of mass of tissue, VD: data is reported in terms of blood velocity, LD: Murray law based data, NP: number of (perforator) arteries supplying the organ. Whenever (×2) is reported, the data is provided for each occurrence. Intra-organ distribution (IOD) is: IE: equally distributed flow through intra-organ delivery vessels, IB: flow distributed following bifurcation law according to lumen radii of intra-organ vessels.

work, more than 2000 specimens were analyzed combining ink injection studies and dissection to create a map of the vascular regions of the body together with the corresponding supplying vessels. That work presented a correlation between the superficial (skin and fascia) and deep (muscles and other tissues) vascular regions with the so-called source arteries, which are those vessels responsible for delivering the blood to the vascular territories through the bifurcation of many perforator arteries. The use of this knowledge has been restricted to clinical applications in the field of reconstructive and plastic surgery, specifically for the design of safe flaps in flap transfer procedures (Saint-Cyr et al, 2008, 2009; Rozen et al, 2010; Sinna et al, 2010; Taylor, 2003; Taylor et al, 2011). After the landmark work of Taylor and Palmer (1987), more advanced techniques have been applied to perform a more accurate description of the vascular territories in the human body, as presented in Chen et al (2009); Morris et al (2006, 2010).

4.2 Modeling hypotheses

In the present work, we employ the conceptual framework presented by (Taylor and Palmer, 1987), as well as the data published in the subsequent years, to define the blood flow distribution throughout the entire human body. Let us describe the way in which we make use of this anatomical data to define our flow distribution criteria.

In deliverying the blood we identify, as already introduced, two classes of arterial vessels closely related to the concept of vascular territory: *source* arteries and *perforator* arteries. Source arteries are medium and small sized arteries responsible for transporting the blood to the peripheral vasculature. Perforator arteries are very small sized arteries which bifurcate from the source arteries, and lead the blood into the tissues through a highly complex and collateralized pre-arteriolar and arteriolar network. Perforator vessels go either through the muscles (musculocutaneous perforators) and through septal spaces into the fascia and skin (septocutaneous perforators) (Taylor, 2003). In the present work we do not distinguish between these classes of perforators, and call these just perforator arteries, or simply perforators. Figure 2 features the ingredients behind the vascular territory concept.

From the vascular point of view, the human body can be understood as a 3D jigsaw puzzle whose pieces are the vascular territories. Under our considerations, these blocks of tissue contain different types of structures, such as muscles, bones, fat, nerves, tendons and skin.

The hypotheses we consider in the use of this concept are the following:

- each vascular territory is supplied just by one source artery or set of functionally similar source arteries;
- the superficial areas of the vascular maps quantitatively reflect what is in the deep layers;

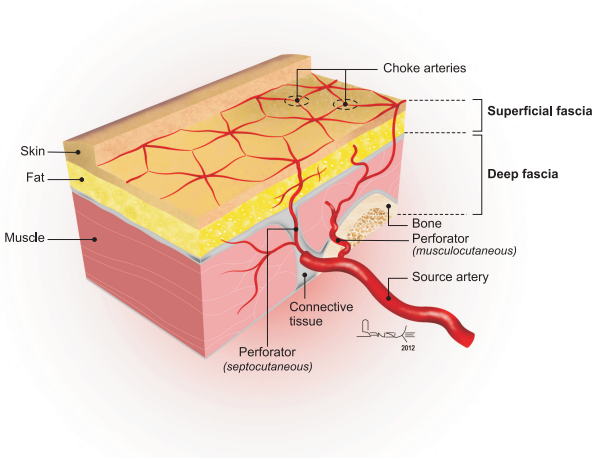


Fig. 2 Schematic view of a source artery feeding a vascular territory through many perforator arteries.

– a vascular territory is essentially a 3D region formed by a mixture of anatomical structures.

The first hypothesis implies that we are neglecting collateralization, which evidently exists, but that in normal conditions plays almost no role. Indeed, the boundaries of vascular territories are actually watershed surfaces which vary dynamically according to the blood microcirculation conditions and to collateral vessels. This collateralization is disregarded in the present work. The second hypothesis is reasonable from the point of view of the proximity with respect to the source artery. In fact, source arteries traverse these blocks (volumes) of tissues delivering blood through the perforator arteries. Accepting that the vascular territories over the surface result from the arrangement of the perforator arteries and the downstream vascular organization, it is reasonable to postulate that the same (or a very similar) vascular organization is taking place in the deeper layers. This hypothesis allows us to convert measures of fractions of area into fractions of volume of tissue through a conversion factor which depends upon the depth of the anatomical structures in the different parts of the human body. The third hypothesis can be circumvented in subsequent studies provided refined data regarding differentiated blood supply to the constituents (muscles, nerves, bones, etc.) is available.

4.3 Blood flow fraction to vascular territories

Let us expose the process to determine the blood flow fraction to each vascular territory in view of the considerations of the previous section.

Table 6 displays global data for the four big regions of the human body, specifically, the number of vascular territories (NT), absolute area (A), regional area fraction (RAF) and number of perforator arteries (NP) are reported. The total area of each one of those four big regions were directly obtained from data reported in Morris et al (2006).

Code	Region	Side	NT	A [cm^2]	RAF (%)	NP
A	Head & Neck	L	10	810	4.91	20
		R	10	810	4.91	20
B	Trunk	–	22	4776	28.98	122
C	Upper Limb	L	16	1671	10.13	57
		R	16	1671	10.13	57
D	Lower Limb	L	21	3375	20.47	92
		R	21	3375	20.47	92
Total			116	16488	100	460

Table 6 Data of vascular territories in the main regions of the human body. L/R: left and right sides, NT: number of vascular territories within the region, A: total area of the region, RAF: regional area fraction, NP: number of perforator arteries within the region.

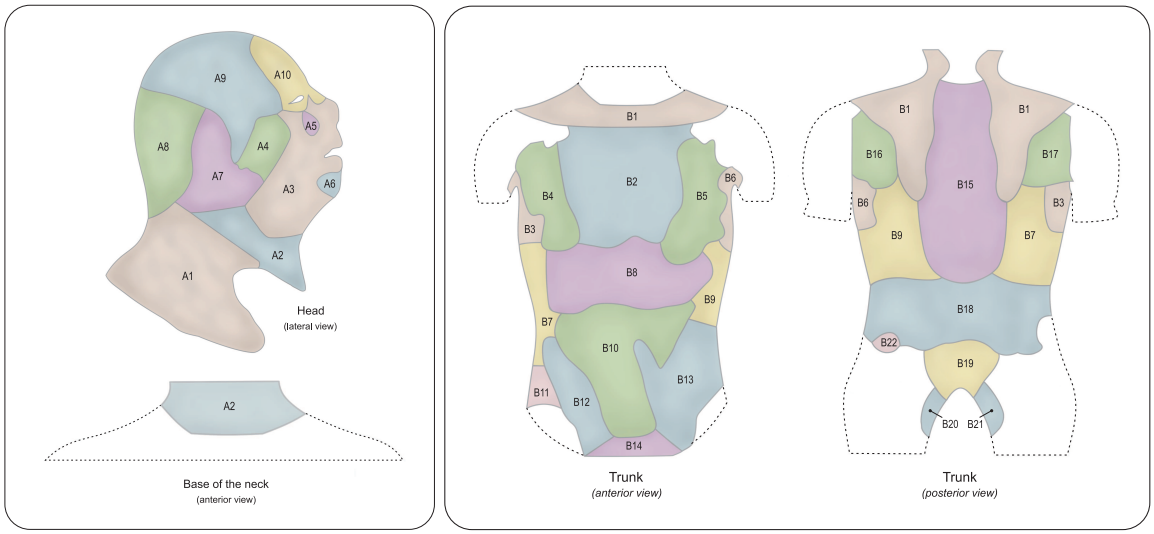
Data in Table 6 is complemented by the segmental division of the four big regions into subregions as shown in Table 7. This subdivision will be explored to determine scaling factors to transform area measures into volume measures, as we will see below.

Code	Region	SR	Subregion
A	Head & Neck	Sc	Scalp
		Ne	Neck
		Fa	Face
B	Trunk	Ch	Chest
		Ab	Abdomen
		Do	Dorsal r.
		Lu	Lumbar r.
C	Upper Limb	Ar	Shoulder & Arm
		Fo	Elbow & Forearm
		Ha	Wrist & Hand
D	Lower Limb	Gl	Gluteal r.
		Th	Hip & Thigh
		Le	Knee & Lower Leg
		An	Ankle & Foot

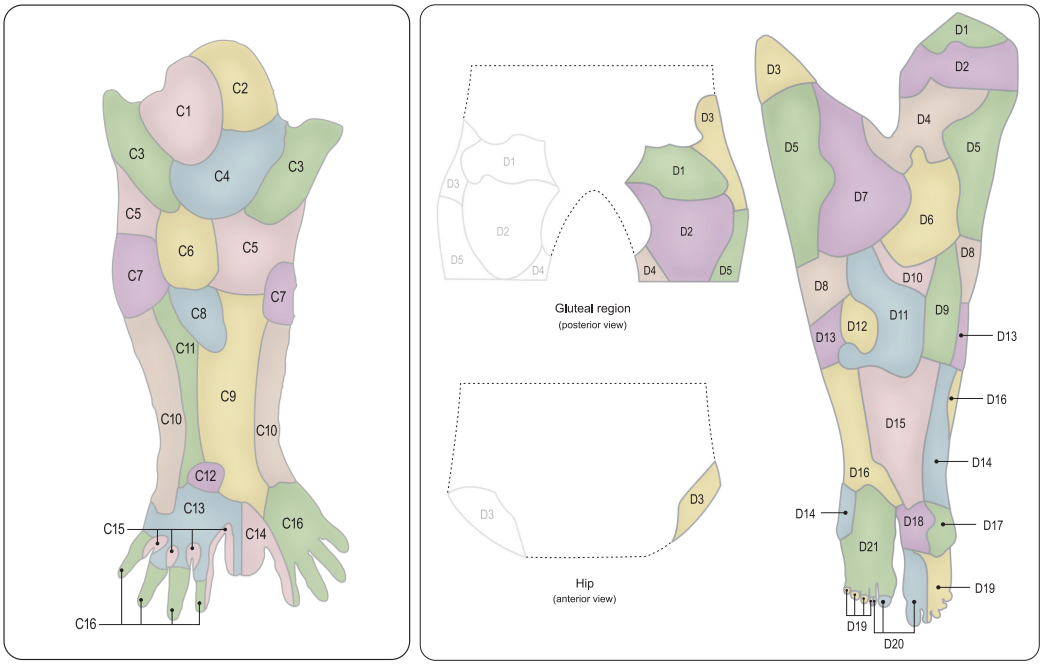
Table 7 Segmental division of regions into subregions of the human body, SR: subregion name, r.: region.

From the data published in Chen et al (2009); Morris et al (2006, 2010), we reproduce an unfolded view of the vascular territories over the surface of the skin for the four big regions of the human body, as seen in Figure 3. The maps of the head, and upper and lower limbs were considered symmetric for simplicity, while the trunk area was treated with some assymetries as reported in the cited works.

The maps from Figure 3 were digitized and used to measure area fractions, which are compiled in Table 8.



(a) Head and neck (b) Trunk



(c) Upper limb (d) Lower limb

Fig. 3 Superficial maps of vascular territories for the regions of the human body. From Chen et al (2009); Morris et al (2010).

The different subregions (see Table 7) are in correspondence with the vascular territories of the human body as specified in Table 8, where the corresponding volume fraction (VF), which is computed through a scaling factor (SF) -see below-, are also reported.

The number and approximate position of the perforator arteries have been taken from Chen et al (2009); Morris et al (2010, 2006); Sauerbier and Unglaub (2010). Such studies provide the average number of perforator arteries which are larger than 0.5 mm in external diameter. This underestimates the real number of perforators because in the real system smaller perforators do exist, but cannot be properly characterized due to their size.

Now, the next step is to transform area fractions of vascular territories into volume fractions. This is carried out by determining a scaling factor which reflects an effective thickness of the anatomical structures. Such scaling factor depends on the subregion of the body (scaling factor for the hand is different from the one of the forearm, and from arm). The segmental laws reported in Pavol et al (2002) allow us to determine the volume fraction for the different subregions (SR) of the

body (hand, forearm, foot, and so on) which are subsets of the four parts. This is employed to compute scaling factors (SF) that yield the volume fractions (VF) according to

$$\mathrm{VF}_{\mathrm{X}Y} = \mathrm{SF}_{\mathrm{X}Y} \times \mathrm{AF}_{\mathrm{X}Y} \quad Y = 1, \dots \mathrm{NT}_{\mathrm{X}} \\ \mathrm{X} \in \{\mathrm{A_s}, \mathrm{B}, \mathrm{C_s}, \mathrm{D_s}\} \quad \mathrm{s} \in \{\mathrm{L}, \mathrm{R}\}, \quad (8)$$

where $\mathrm{s} \in \{\mathrm{L}, \mathrm{R}\}$, (L: left, R: right). The so computed volume fractions, and the corresponding scaling factors obtained from the mass fractions reported in Pavol et al (2002) (assuming constant density), are given in Table 8.

In obtaining the final blood flow fraction (BFF) also reported in Table 8 we need to further process the volume fraction (VF) to accommodate the known flow fraction supplied as a whole to each region. This information was retrieved from the data published in Ahlborg and Jensen-Urstad (1991); Isnard et al (1996) for arms (measuring subclavian venous return) and legs (measuring femoral venous return), respectively, and from Sato et al (2011) for the head (measuring external carotid arterial blood flow). The flow fraction to the trunk is the remaining part. These data are called target flow fractions (TFF), and are compiled in Table 9. Such table also contains the total volume corresponding to each region (RVF), which is obtained as

$$\mathrm{RVF}_{\mathrm{X}} = \sum_{Y=1}^{\mathrm{NT_X}} \mathrm{VF}_{\mathrm{X}Y} \\ \mathrm{X} \in \{\mathrm{A_s}, \mathrm{B}, \mathrm{C_s}, \mathrm{D_s}\} \quad \mathrm{s} \in \{\mathrm{L}, \mathrm{R}\}, \quad (9)$$

Code	Region	Side	RVF (%)	TFF (%)
A	Head & Neck	L	2.4475	2
		R	2.4475	2
B	Trunk	–	28.9120	7.1915
C	Upper Limb	L	5.8828	3.5
		R	5.8828	3.5
D	Lower Limb	L	27.2137	8.5
		R	27.2137	8.5
Total			100	35.1915

Table 9 Global blood flow distribution to vascular territories. L/R: left and right sides, RVF: regional volume fraction (according to data from Table 8, see equation (9)), TFF: target blood flow fraction according to published data.

Finally, the blood flow fraction (BFF) to each territory is obtained as follows. On the one hand, in Cormack and Lamberty (1986), a curve-fitting process on measurements in sheet-like tissues showed a cubic relation between the perfused area of territories and the radius of source arteries. Assuming constant thickness of the structures, it is possible to reach a proportional relation between the perfused volume and the vessel radius to the third power. Hence, considering Murray's law (Murray, 1926) we reach a linear relation between flow and volume. On the other hand, studies of allometric laws established that this relation should follow a power law with power equal to 3/4 (West et al, 1997). In any case, we generalize such relation using the generic power law

$$\mathrm{BFF}_{\mathrm{X}Y} = \sigma \left[\frac{\mathrm{TFF_X}}{\mathrm{RVF_X}} \mathrm{VF}_{\mathrm{X}Y} \right]^{\gamma} \quad Y = 1, \dots \mathrm{NT_X} \\ \mathrm{X} \in \{\mathrm{A_s}, \mathrm{B}, \mathrm{C_s}, \mathrm{D_s}\} \quad \mathrm{s} \in \{\mathrm{L}, \mathrm{R}\}. \quad (10)$$

with

$$\sigma = \frac{\sum_{\mathrm{X}} \mathrm{TFF_X}}{\sum_{\mathrm{X}} \sum_{Y=1}^{\mathrm{NT_X}} \left[\frac{\mathrm{TFF_X}}{\mathrm{RVF_X}} \mathrm{VF}_{\mathrm{X}Y} \right]^{\gamma}} \quad (11)$$

where $\sum_{\mathrm{X}}$ is over the set $\{\mathrm{A_s}, \mathrm{B}, \mathrm{C_s}, \mathrm{D_s}\}$, $\mathrm{s} \in \{\mathrm{L}, \mathrm{R}\}$. Expression (10) states that the blood flow fraction in a given territory follows a power law with respect to the volume fraction of such territory (given by (8)) affected by a scalar factor that takes into account a relative weigh of the amount of flow delivered to the region the territory belongs to and the volume of that region. The role of σ is to scale the flow fractions such that the sum of the BFF is the total TFF (see Table 8).

These are the ultimate data we need to determine the blood supply to the vascular territories. Choosing between $\gamma = 1$ and $\gamma = 3/4$ we are in the aforementioned cases. For simplicity, in the present work we consider $\gamma = 1$, for which the values of BFF are also reported in Table 8. Preliminary studies show that results are qualitatively equivalent for both choices of γ. A deeper quantitative comparison is current of research.

5 Calibration algorithm

To satisfy the blood flow distribution to specific organs (Section 3) and distributed organs (Section 4), we extend to the whole arterial network the methodology proposed for the arm in (Blanco et al, 2012b). We present two formulations to be compared: a simple problem called *simple analytical approach* (SAA), which is valid for the inviscid limit, and an estimation approach, called *parameter identification problem* (PIP).

VT	Source artery	A [cm^2]	RAF (%)	AF (%)	NP	SR	SF	VF (%)	BFF (%)
A1	thyrocervical t.	178.6	22.0	1.083	3	Ne	0.498	0.5396	0.4409
A2	superior thyroid a.	186.8	23.1	1.133	4	Ne	0.498	0.5643	0.4611
A3	facial a.	100.7	12.4	0.611	2	Fa	0.498	0.3043	0.2487
A4	transverse facial a.	28.6	3.5	0.174	1	Fa	0.498	0.0865	0.0707
A5	infraorbital a.	3.9	0.5	0.024	1	Fa	0.498	0.0117	0.0096
A6	mental a.	6.2	0.8	0.037	1	Fa	0.498	0.0186	0.0152
A7	posterior auricular a.	65.2	8.1	0.396	2	Sc	0.498	0.1971	0.1611
A8	occipital a.	83.4	10.3	0.506	2	Sc	0.498	0.2521	0.2060
A9	superficial temporal a.	122.3	15.1	0.742	3	Sc	0.498	0.3696	0.3020
A10	ophthalmic a.	34.3	4.2	0.208	1	Fa	0.498	0.1036	0.0847
B1	thyrocervical t.	613.8	12.8	3.722	16	Ch-Do	0.998	3.7155	0.9242
B2	internal thoracic a.	530.6	11.1	3.217	12	Ch	0.998	3.2121	0.7990
B3	right thoracodorsal a.	81.6	1.7	0.495	2	Do	0.998	0.4938	0.1228
B4	right lateral thoracic	169.7	3.6	1.029	4	Ch	0.998	1.0271	0.2555
B5	left lateral thoracic a.	214.2	4.5	1.299	5	Ch	0.998	1.2967	0.3225
B6	left thoracodorsal a.	74.7	1.6	0.453	2	Do	0.998	0.4520	0.1124
B7	lateral bb. of right posterior intercostal aa.	255.4	5.3	1.549	6	Do	0.998	1.5463	0.3846
B8	superior epigastric a.	375.5	7.9	2.278	10	Ab	0.998	2.2734	0.5655
B9	lateral bb. of left posterior intercostal aa.	253.1	5.3	1.535	6	Do	0.998	1.5321	0.3811
B10	deep inferior epigastric a.	419.2	8.8	2.543	12	Ab	0.998	2.5379	0.6313
B11	right deep circumflex iliac a.	37.2	0.8	0.226	1	Ab	0.998	0.2252	0.0560
B12	right superficial inferior epigastric a.	144.1	3.0	0.874	4	Ab	0.998	0.8724	0.2170
B13	left superficial inferior epigastric a.	274.2	5.7	1.663	7	Ab	0.998	1.6601	0.4129
B14	superficial external pudendal a.	62.5	1.3	0.379	2	Ab	0.998	0.3781	0.0941
B15	dorsal bb. of posterior intercostal aa.	490.2	10.3	2.972	12	Do	0.998	2.9675	0.7381
B16	left circumflex scapular a.	95.9	2.0	0.582	2	Do	0.998	0.5808	0.1445
B17	right circumflex scapular a.	93.7	2.0	0.568	2	Do	0.998	0.5672	0.1411
B18	lumbar aa.	412.5	8.6	2.501	10	Lu	0.998	2.4973	0.6212
B19	lateral sacral aa.	111.4	2.3	0.675	4	Lu	0.998	0.6741	0.1677
B20	left internal pudendal a.	22.1	0.5	0.134	1	Lu	0.998	0.1336	0.0332
B21	right internal pudendal a.	29.8	0.6	0.181	1	Lu	0.998	0.1804	0.0449
B22	left deep circumflex iliac a.	14.6	0.3	0.089	1	Lu	0.998	0.0885	0.0220
C1	posterior humeral circumflex a.	119.6	7.2	0.725	3	Ar	0.643	0.4666	0.2776
C2	thoracoacromial a.	88.0	5.3	0.534	3	Ar	0.643	0.3433	0.2043
C3	superior ulnar collateral a.	181.4	10.9	1.100	2	Ar	0.643	0.7078	0.4211
C4	deep braquial a.	129.2	7.7	0.783	4	Ar	0.643	0.5038	0.2997
C5	braquial a.	140.2	8.4	0.851	6	Ar	0.643	0.5470	0.3255
C6	radial collateral a.	79.1	4.7	0.480	2	Ar	0.643	0.3087	0.1837
C7	inferior ulnar collateral a.	99.0	5.9	0.601	2	Ar	0.643	0.3862	0.2298
C8	radial recurrent a.	48.5	2.9	0.294	2	Fo	0.608	0.1790	0.1065
C9	radial a.	199.5	11.9	1.209	5	Fo	0.608	0.7358	0.4378
C10	ulnar a.	188.8	11.3	1.145	7	Fo	0.608	0.6967	0.4145
C11	posterior interosseous a.	64.6	3.9	0.392	5	Fo	0.608	0.2383	0.1418
C12	anterior interosseous a.	16.3	1.0	0.099	3	Fo	0.608	0.0603	0.0359
C13	dorsal metacarpal aa.	97.6	5.8	0.592	4	Ha	0.369	0.2185	0.1300
C14	princeps pollicis a.	41.9	2.5	0.254	2	Ha	0.369	0.0938	0.0558
C15	common palmar digital a.	30.2	1.8	0.183	1	Ha	0.369	0.0677	0.0403
C16	superficial palmar arch	147.1	8.8	0.892	6	Ha	0.369	0.3294	0.1960
D1	superior gluteal a.	135.5	4.0	0.822	4	Gl	1.800	1.4787	0.4618
D2	inferior gluteal a.	302.4	9.0	1.834	8	Gl	1.800	3.3009	1.0310
D3	superficial circumflex iliac a.	160.9	4.8	0.976	4	Gl	1.800	1.7568	0.5487
D4	medial circumflex femoral a.	199.5	5.9	1.210	5	Gl	1.800	2.1775	0.6801
D5	lateral circumflex femoral a.	502.7	14.8	3.048	14	Th	1.974	6.0179	1.8796
D6	deep femoral a.	199.1	5.9	1.208	5	Th	1.974	2.3836	0.7445
D7	femoral a.	325.6	9.6	1.975	9	Th	1.974	3.8990	1.2178
D8	superior lateral genicular a.	127.9	3.8	0.776	4	Le	0.667	0.5177	0.1617
D9	popliteal a.	118.3	3.5	0.718	3	Le	0.667	0.4789	0.1496
D10	superior medial genicular a.	48.2	1.4	0.292	1	Le	0.667	0.1949	0.0609
D11	descending genicular a.	221.5	6.6	1.343	6	Le	0.667	0.8965	0.2800
D12	inferior medial genicular a.	54.8	1.6	0.333	2	Le	0.667	0.2219	0.0693
D13	inferior lateral genicular a.	73.2	2.2	0.444	2	Le	0.667	0.2963	0.0925
D14	fibular a.	123.9	3.7	0.752	3	Le	0.667	0.5017	0.1567
D15	posterior tibial a.	255.0	7.6	1.547	7	Le	0.667	1.0322	0.3224
D16	anterior tibial a.	166.3	4.9	1.008	5	Le	0.667	0.6731	0.2102
D17	calcaneal b. of posterior tibial a.	38.4	1.1	0.233	1	An	0.635	0.1477	0.0461
D18	calcaneal b. of fibular a.	48.1	1.4	0.292	1	An	0.635	0.1853	0.0579
D19	lateral plantar a.	56.1	1.7	0.340	2	An	0.635	0.2158	0.0674
D20	medial plantar a.	56.9	1.7	0.345	2	An	0.635	0.2189	0.0684
D21	dorsalis pedis a.	160.7	4.8	0.974	4	An	0.635	0.6184	0.1932
Total*		16488		100	460			100	35.1915

Table 8 Data for the vascular territories of the regions of the body. VT: code for the vascular territory, A: area of the territory, RAF: regional area fraction, AF: body area fraction, NP: number of perforator arteries to the territory, SF: scaling factor, VF: body volume fraction (see equation (8)), BFF: blood flow fraction (obtained using data from Table 9, see equation (10), for $\gamma = 1$), SR: subregion containing territory (see Table 7), a.: artery, t. trunk, b: branch, aa.: arteries, bb: branches. *: values of regions A, C and D have to be doubled (left and right sides).

5.1 Parameter identification problem (PIP)

Overall, there are 144 blood flow destinations: $N_{\mathrm{O}} = 28$ specific organs and $N_{\mathrm{T}} = 116$ vascular territories. For a given cardiac output (CO), with the value of BFF reported in Table 5 and in Table 8 we obtain the averaged flow data to each vascular destination

$$\mathbf{Q} = \big(\{\bar{Q}_{\mathrm{T}Y}\}_{Y=1}^{N_{\mathrm{T}}}, \{\bar{Q}_{\mathrm{O}Y}\}_{Y=1}^{N_{\mathrm{O}}}\big). \tag{12}$$

where the code T groups all the N_{T} vascular territories, and O the N_{O} specific organs. From Table 8 it can be observed that

$$\{\bar{Q}_{\mathrm{T}Y}\}_{Y=1}^{\mathrm{N_T}} = \bigcup_{\substack{\mathrm{X}\in\{\mathrm{A_s},\mathrm{B},\mathrm{C_s},\mathrm{D_s}\}\\ \mathrm{s}\in\{\mathrm{L},\mathrm{R}\}}} \{\bar{Q}_{\mathrm{X}Y}\}_{Y=1}^{\mathrm{NT_X}}. \tag{13}$$

For ease of notation we hereafter continue using the following denomination $\mathbf{Q} = (\bar{Q}_1, \ldots, \bar{Q}_N)$, $N = N_{\mathrm{T}} + N_{\mathrm{O}} = 144$. This indexing matches the ordering established in expressions (12)-(13), and $\bar{(\cdot)}$ denotes averaged quantities over a heart beat. The resistances corresponding to the specific organs and to the vascular territories are therefore denoted by $\mathbf{R} = (R_1, \ldots, R_N)$.

The one-dimensional model, described through equations (1)-(6), is an operator which, for a given set of resistances $\mathbf{R}'$, renders a certain flow over each of the 144 destinations. This outcome is denoted by $\boldsymbol{\mathcal{Q}}(\mathbf{R}') = (\bar{\mathcal{Q}}_1(\mathbf{R}'), \ldots, \bar{\mathcal{Q}}_N(\mathbf{R}'))$. Then, one should seek for the set of resistances such that

$$\mathbf{Q} = \boldsymbol{\mathcal{Q}}(\mathbf{R}). \tag{14}$$

Nevertheless, since we are imposing a certain inflow condition $Q_{\mathrm{in}}(t)$ at the aortic root, with an average $\bar{Q}_{\mathrm{in}}$, and since we have mass conservation over the cardiac period (once the system is in the periodic state), the following conservation equation is satisfied

$$\bar{Q}_{\mathrm{in}} = \sum_{i=1}^{N} \bar{\mathcal{Q}}_i(\mathbf{R}'). \tag{15}$$

Hence, problem posed by equation (14) is overdetermined in the flow unknowns. On the other hand, averaged pressure is not determined by the system. Therefore, we proceed by replacing one of the equations in (14) (say equation m) by the following equation that states that, for resistance $\mathbf{R}'$, the averaged pressure at the inlet point of the network (the aortic root), denoted by $\bar{\mathcal{P}}_{\mathrm{in}}(\mathbf{R}')$ should equal a given datum $\bar{P}_{\mathrm{in}}$. Then, we arrive at the formulation of the identification problem.

Parameter identification problem: for a given datum $\mathbf{D} = (\bar{Q}_1, \ldots, \bar{P}_{\mathrm{in}}, \ldots, \bar{Q}_N)$, find the set of resistances $\mathbf{R} = (R_1, \ldots, R_N)$ such that

$$\mathbf{D} = \boldsymbol{\mathcal{M}}(\mathbf{R}), \tag{16}$$

where $\boldsymbol{\mathcal{M}}(\mathbf{R}) = (\bar{\mathcal{Q}}_1(\mathbf{R}), \ldots, \bar{\mathcal{P}}_{\mathrm{in}}(\mathbf{R}), \ldots, \bar{\mathcal{Q}}_N(\mathbf{R}))$ is the so-called *model operator*.

The model operator introduced above is a non-linear operator which amounts to solving, for a given set of resistances, the equations of the 1D model presented in Section 2.2 over the arterial network described in Section 2.1, and then computing the averaged quantities at the corresponding inlet/outlet points throughout the system. It is worthwhile to point out that the fully periodic state has to be reached to evaluate the outcome of the operator.

Equation (16) is solved using the Newton method. The residual is defined as

$$\mathbf{r}(\mathbf{R}') = \boldsymbol{\mathcal{M}}(\mathbf{R}') - \mathbf{D}, \tag{17}$$

and the Jacobian is

$$\mathbf{J}(\mathbf{R}') = \left.\frac{\partial \mathbf{r}}{\partial \mathbf{R}}\right|_{\mathbf{R}'} = \left.\frac{\partial \boldsymbol{\mathcal{M}}}{\partial \mathbf{R}}\right|_{\mathbf{R}'}, \tag{18}$$

Then, the algorithm reads:

1. Given $\mathbf{R}^0$
2. Compute $\mathbf{r}^0 = \mathbf{r}(\mathbf{R}^0)$ and $\mathbf{J}^0 = \frac{\partial \mathbf{r}}{\partial \mathbf{R}}(\mathbf{R}^0)$
3. For $k = 0, 1, \ldots$ until $\frac{\|\mathbf{r}^k\|}{\|\mathbf{r}^0\|} \leq \varepsilon$ do
   4. $\Delta\mathbf{R}^j = -(\mathbf{J}^k)^{-1}\mathbf{r}^k$
   5. $\mathbf{R}^{k+1} = \mathbf{R}^k + \Delta\mathbf{R}^k$
   6. $\mathbf{r}^{k+1} = \mathbf{r}(\mathbf{R}^{k+1})$
   7. $\mathbf{J}^{k+1} = \frac{\partial \mathbf{r}}{\partial \mathbf{R}}(\mathbf{R}^{k+1})$
8. Enddo

In view of the complexity of the Jacobian we compute $\mathbf{J}(\mathbf{R}^k)$ by finite differences. For a given m, $j = 1, \ldots, N$ and $i = 1, \ldots, N$, $i \neq m$, the components of the Jacobian are

$$[\mathbf{J}(\mathbf{R}^k)]_{ij} = \frac{\bar{\mathcal{Q}}_i(\mathbf{R}^k + \delta R^k \mathbf{e}_j) - \bar{\mathcal{Q}}_i(\mathbf{R}^k)}{\delta R^k} \qquad i, j = 1, \ldots, N \quad i \neq m, \tag{19a}$$

$$[\mathbf{J}(\mathbf{R}^k)]_{mj} = \frac{\bar{\mathcal{P}}_{\mathrm{in}}(\mathbf{R}^k + \delta R^k \mathbf{e}_j) - \bar{\mathcal{P}}_{\mathrm{in}}(\mathbf{R}^k)}{\delta R^k} \qquad j = 1, \ldots, N, \tag{19b}$$

where $\mathbf{e}_j$ is the canonical vector with unit value in the jth position and zero elsewhere, and $\delta R^k = \tau \|\mathbf{R}^k\|$ with τ chosen sufficiently small. Hereby, the pair $(\mathbf{R}, \boldsymbol{\mathcal{Q}}(\mathbf{R}))$ is called *PIP solution*.

5.2 Intra-organ flow distribution

As noted, we are addressing the blood flow supply to the different vascular regions, both specific organs and vascular territories, but nothing has been said regarding the intra-organ distribution. According to Table 5 and Table 8, the number of perforator arteries which supply each vascular region is larger than one ($\mathrm{NP} \geq 1$). In the case of specific organs, the intra-organ distribution is addressed as detailed in Table 5, by considering an equally distributed flow through the perforators or by considering the Murray law to derive the flow within the organ. In the case of vascular territories, in all cases we consider an equally distributed flow through the perforators.

Then, the 144 resistances that are the input for the model operator $\boldsymbol{\mathcal{M}}$ in the evaluation of the residual (17) have to be mapped into the 988 arteries that ultimately supply the blood to terminal locations (528 perforator arteries and arterial segments to specific organs and 460 perforator arteries to vascular territories). This step is necessary at each Newton iteration, and is part of the internal definition of the model operator.

For those specific organs for which flow is distributed equally through the different perforator arteries we have

$$R_{\mathrm{O}Y,T_j} = \frac{R_{\mathrm{O}Y}}{\mathrm{NP}_{\mathrm{O}Y}} \quad j = 1, \ldots, \mathrm{NP}_{\mathrm{O}Y} \quad Y \in \mathcal{Y}_{\mathrm{IE}}, \tag{20}$$

where $\mathcal{Y}_{\mathrm{IE}}$ is the set of indexes of specific organs which have the IE denomination in Table 5. For those organs which follow the Murray law we have

$$R_{\mathrm{O}Y,T_j} = \frac{1}{r^3_{\mathrm{O}Y,T_j}} \left(\sum_{i=1}^{\mathrm{NP}_{\mathrm{O}Y}} r^3_{\mathrm{O}Y,T_i} \right) R_{\mathrm{O}Y} \quad j = 1, \ldots, \mathrm{NP}_{\mathrm{O}Y} \quad Y \in \mathcal{Y}_{\mathrm{IB}}, \tag{21}$$

where $r_{\mathrm{O}Y,T_j}$ is the lumen radius of the arterial segment that reaches the terminal location j in the organ $\mathrm{O}Y$, and $\mathcal{Y}_{\mathrm{IB}}$ is the set of indexes of specific organs with IB denomination in Table 5.

For all vascular territories we simply have

$$R_{\mathrm{X}Y,T_j} = \frac{R_{\mathrm{X}Y}}{\mathrm{NP}_{\mathrm{X}Y}} \quad j = 1, \ldots, \mathrm{NP}_{\mathrm{X}Y} \quad Y = 1, \ldots, \mathrm{NT}_{\mathrm{X}} \quad \mathrm{X} \in \{\mathrm{A_s}, \mathrm{B}, \mathrm{C_s}, \mathrm{D_s}\} \quad \mathrm{s} \in \{\mathrm{L}, \mathrm{R}\}. \tag{22}$$

5.3 Terminal compliance

Peripheral compliance is taken proportional to blood flow fraction, as proposed in Stergiopulos et al (1992). For this, we estimate the peripheral compliance (at terminals) as

$$C_{\mathrm{per}} = \theta_C C_{T,\mathrm{lin}}, \tag{23}$$

with $\theta_C = 0.1$, where $C_{T,\mathrm{lin}}$ is the total compliance of the network considering just the linear contribution of the elastin stiffness E_E. This simplification is reasonable because the elastin is responsible for bearing approximately 80% of the pressure load in normal conditions, dominating the compliance of the tree. The value for θ_C has been taken arbitrarily small. In fact, as will be shown in the sensitivity analysis, the peripheral compliance do not influence the result at large and medium-sized arteries. As a reference value it is $\theta_C = 0.16$ in Stergiopulos et al (1992), while in our model, we took $\theta_C = 0.1$ because of the larger number of vessels that compose our network. A linear analysis of the system can yield the peripheral compliance from the difference of the total network compliance and the compliance of 1D segments in the tree (Alastruey et al, 2008b).

Hence, at each terminal it is

$$C_k = \mathrm{BFF}_k C_{\mathrm{per}} \quad \forall k, \tag{24}$$

where BFF_k is the blood flow fraction through terminal k, which is an index that ranges over all terminal locations in the tree. In this work it is $C_{\mathrm{per}} = 9.5274 \cdot 10^{-5}$ cm^5/dyn.

5.4 Terminal pressure

At terminal locations pressure level should match the venous pressure. This low pressure is, in normal conditions, negligible with respect to the arterial pressure, and acts actually as a reference pressure for the model. This is why in the present work we put this reference value to zero by taking null pressure at terminal locations (except those terminals corresponding to the coronary tree which supply the heart -organ O1 in Table 5-), that is

$$P_{\mathrm{O}Y,T_j} = 0 \quad j = 1, \ldots, \mathrm{NP}_{\mathrm{O}Y} \quad Y = 2, \ldots, N_{\mathrm{O}}, \tag{25}$$

and

$$P_{\mathrm{X}Y,T_j} = 0 \quad j = 1, \ldots, \mathrm{NP}_{\mathrm{X}Y} \quad Y = 1, \ldots, \mathrm{NT}_{\mathrm{X}}. \tag{26}$$

For the coronary tree a time-dependent terminal pressure P_h is considered to reproduce the conditions that lead to diastolic fill-in of the arteries in the heart.

$$P_{\mathrm{O1},T_j} = P_{T,h}(t) \quad j = 1, \ldots, \mathrm{NP}_{\mathrm{O1}}. \tag{27}$$

Here, $P_{T,h}(t)$ is taken to be 60% of the left ventricle pressure (Bovendeerd et al, 2006), and is shown in Figure 4.

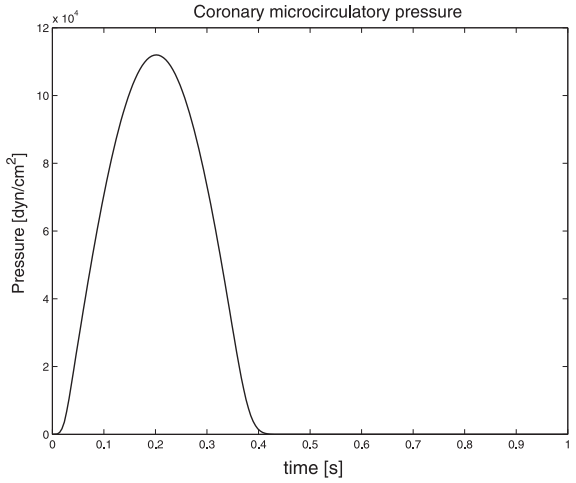


Fig. 4 Terminal pressure $P_{T,h}(t)$ at the coronary tree.

5.5 Simple analytical approach (SAA)

In the inviscid limit equation (16) is analytically solvable. This analytical solution, denoted by $\mathbf{R}^{\mathrm{a}}$, is

$$\mathbf{R}^{\mathrm{a}} = (R_1^{\mathrm{a}}, \ldots, R_N^{\mathrm{a}}) = \left(\frac{\Delta \bar{P}_1}{\bar{Q}_1}, \ldots, \frac{\Delta \bar{P}_N}{\bar{Q}_N} \right). \tag{28}$$

where $\Delta \bar{P}_i = \bar{P}_{\mathrm{in}} - \bar{P}_{T,i}$, $i = 1, \ldots, N$, with $\bar{P}_{T,i}$ the average value of the terminal pressure as given in Section 5.4. The pair $(\mathbf{R}^{\mathrm{a}}, \boldsymbol{\mathcal{Q}}(\mathbf{R}^{\mathrm{a}}))$ is called *SAA solution.*

5.6 Numerical method for blood flow equations

Equations (1)-(2) are solved using a least squares finite element approach for the characteristic equations (see Watanabe et al (2013)). Such approach consists of a first order finite difference method in time and a first order Lagrange polynomial approximation in space. The wall equation (3) is manipulated using a first order finite difference method to split the elastic and viscoelastic parts. This results in a fully implicit non-linear system of algebraic equations which is linearized using fixed point iterations.

Convergence studies were performed. The chosen time step is $\Delta t = 0.0025$ s, and the space discretization is given by $\Delta x = 0.5$ cm, resulting in 122,000 degrees of freedom. Arterial segments are composed of at least four elements. Since we use an implicit method in time, no CFL condition must be fulfilled.

5.7 Parallel implementation

The computation of the Jacobian through finite differences turns to be the best approach in view of the complexity of an analytical calculation of the derivatives. However, this poses a great challenge from the point of view of the computational cost. The Jacobian evaluation through (19a)-(19b) requires $N + 1$ evaluations of the residual, which amounts to solving $N + 1$ the 1D model in such a complex topology during several cardiac cycles to guarantee the periodic state, and so compute the averaged quantities which are the outcome of the model operator. A single evaluation (10 heart beats) could take up two hours, which would inviabilize the present approach. This motivated a fully parallel implementation for the computation of the derivatives. The assembly of the Jacobian in Step 7 of the Newton algorithm described before is an embarrassingly parallel procedure, and therefore fully scalable.

6 Numerical experiments

6.1 Computational issues and numerical setting

To ensure the periodic state we simulate 10 cardiac cycles in the evaluation of the model operator. The model equations are solved in a sequential manner, while many instances run in parallel as explained in Section 5.7. Each Newton iteration consists of those many instances running in parallel each of which performs 10 cardiac cycles. Each cardiac cycle takes approximately 11 mins. to be solved in a Intel® Xeon® Processor E5-2650 running at 2.00GHz. The convergence tolerance of the Newton method is $\epsilon = 1 \cdot 10^{-4}$ and the perturbation $\tau = \epsilon = 1 \cdot 10^{-4}$. This perturbation parameter depends on the convergence tolerance in a non-trivial manner and has been manually adjusted. In solving equation (16), the Newton method is initialized with the SAA solution, i.e. $\mathbf{R}^0 = \mathbf{R}^{\mathrm{a}}$. The equation that is removed in writing (16) from (14) corresponds to one of the kidneys, for which it is $m = 130$. The full calibration procedure takes 7 Newton iterations, yielding a total calibration time of approximately 13 hrs..

The arterial flow is driven by an inflow boundary condition at the aortic root given by the flow curve shown in Figure 5 obtained from Murgo et al (1980), for which the cardiac output (CO) is $\bar{Q}_{\mathrm{in}} = 6.727$ lt/min, with a cardiac period $T = 1$ s.

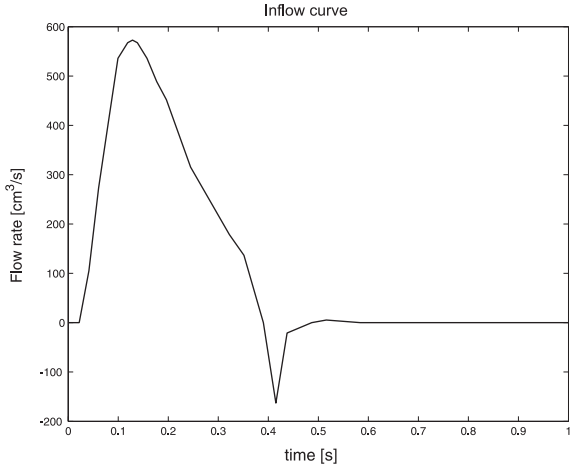


Fig. 5 Inflow condition in the arterial network.

6.2 Model predictions

Overall, pressure and flow rate waveforms as predicted by the ADAN model are in well agreement with data obtained from measurements as well as with other computational models. Qualitative, and in some cases quantitative, comparisons of pressure and flow waveforms with data published in the literature are given in Figure 6 and Figure 7, respectively.

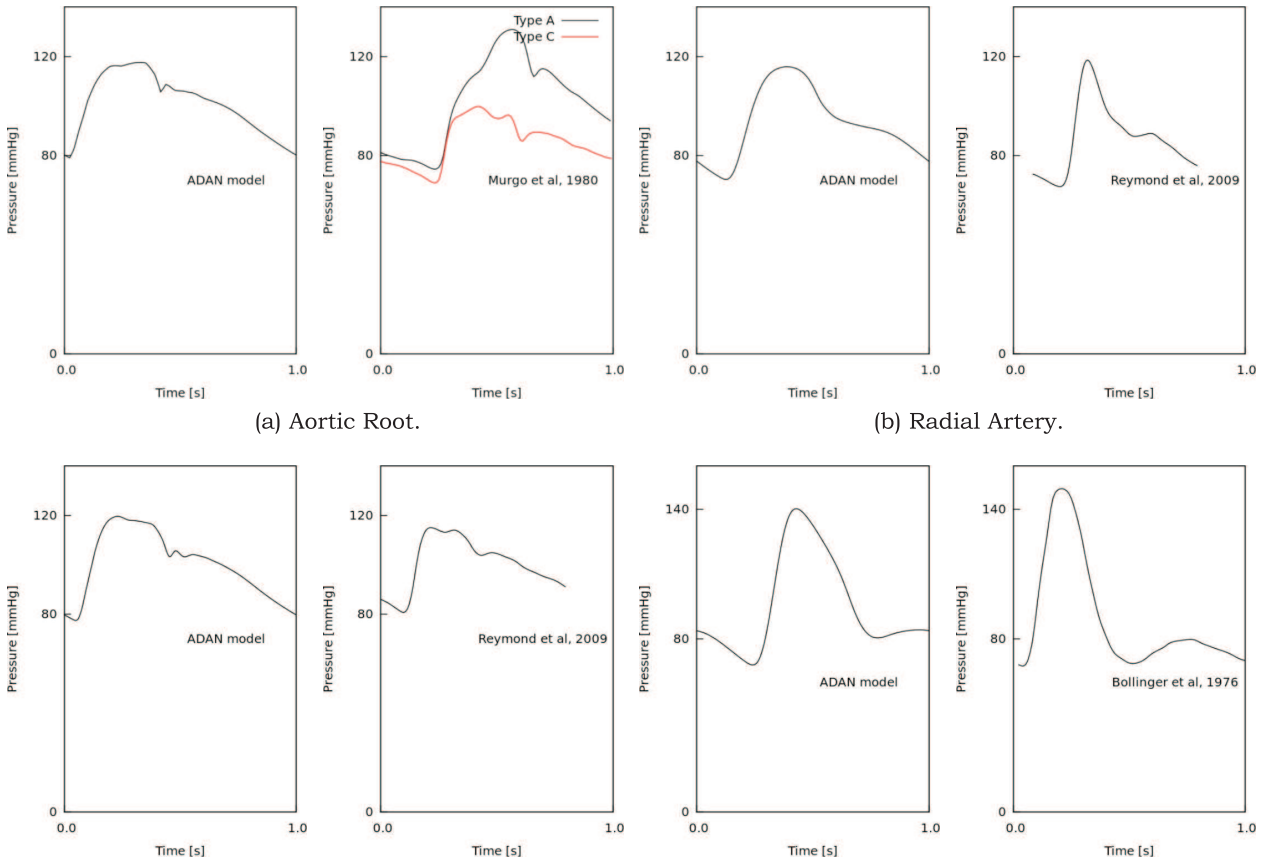
The pressure pulse at the aortic root is a typical contour corresponding to a healthy young subject, type B curve (see Figure 6(a), and the data extracted from Murgo et al (1980) -Figures 1,3-). These aortic root pressure curves are characterized in terms of the amplification index (AI, see explanation below): AI > 12 is type A, $0 <$ AI < 12 is type B and AI < 0 is type C. For the current model it is AI $= 3.7$ (see Table 10). The salient feature of this kind of contour is the concavity during diastole (Vlachopoulos and O'Rourke, 2000). Radial artery and common carotid pressure waveforms are very similar to measurements published in Reymond et al (2009) (see Figure 6(b) and Figure 6(c)). Also, the pressure waveform predicted at the posterior tibial artery is in agreement measurements published in Bollinger et al (1976) (see Figure 6(d)).

At the abdominal aorta the blood flow has a shift when going from proximal to distal locations. A biphasic waveform with flow reversal in early diastole is obtained in the proximal abdominal aorta (supraceliac location) and a triphasic waveform with forward flow in late diastole is observed in the distal part of the abdominal aorta (infrarenal location). This is in agreement with measurements reported in (Bhatt et al, 2007) (see Figure 7(a) and Figure 7(b)). In turn, the flow waveform at the thoracic aorta agrees with measurements reported in Reymond et al (2009) (see Figure 7(c)). In the lower limb, waveform predictions at the femoral artery correspond to a typical flow contour also as reported in Reymond et al (2009) (see Figure 7(d)). Furthermore, when comparing to waveforms reported in Reymond et al (2009), the present model yields characteristic flow rate contours at the common carotid artery and at the middle cerebral artery (see Figure 7(e) and Figure 7(f)). At the basilar artery the results agree with measurements published in Gwilliam et al (2009) (see Figure 7(g)). Regarding coronary circulation, as expected the model predicts diastolic fill-in (see Figure 7(h)). Moreover, the flow contour exhibits oscillating behavior as does data reported in Johnson et al (2008). Note that measurements in subjects show a large interindividual variability. These data were not used in the calibration of the ADAN model, so the results delivered by the ADAN model are true prediction. Although the velocities are higher in our model, the flow waveform resembles that of subject B from Figure 7(h).

Modification of the pressure pulse as it travels along the aorta towards the peripheral beds also presents characteristic features as seen in Figure 8. This is in fair agreement with the domain knowledge (see Olufsen et al (2000)). In the same figure, flow contours provide an account of the hemodynamics taking place in very specific, sometimes distant, arterial locations throughout the entire network. A more detailed analysis of these results will be presented in a forthcoming contribution.

In addition, a quantitative comparison of most important cardiovascular indexes is presented in Table 10. In that table:

- AI is the amplification index, which is defined as $\text{AI} = \frac{P_2 - P_1}{\Delta P} 100$, where P_2 and P_1 are, respectively, the maximum pressure peak and the pressure at the previous inflection point during systole, and ΔP is the pulse pressure (systolic minus diastolic pressure);
- ABI is the ankle-brachial index computed as $\text{ABI} = \frac{P_{\text{arm}}}{P_{\text{leg}}}$, where P_{arm} is the systolic pressure in the arm (brachial artery usually) and P_{leg} is the systolic pressure in the leg at the level of the ankle (posterior tibial artery or dorsalis pedis artery, the largest pressure recorded);
- PWV is the pulse wave velocity which is computed as $\text{PWV}_{\text{XY}} = \frac{\Delta L_{\text{XY}}}{\Delta T_{\text{XY}}}$, where ΔL_{XY} is the distance from aortic root to distal site at vessel Y minus the distance from aortic root to proximal site at vessel X, and ΔT_{XY} is the foot-to-foot time in pressure waveforms.



(a) Aortic Root. (b) Radial Artery.

(c) Common Carotid Artery. (d) Posterior Tibial Artery.

Fig. 6 Comparison of pressure waveforms between the ADAN model and published data.

Index	ADAN model	Ref. values
HR [bpm]	60	81[1]
CO [lt/min]	6.73	7.01[1]
(DBP-SBP) [mmHg]	(79-118)	(80-114)[1]
AI_A	3.77	[0,15][3]
AI_C	-6.93	[-20,0][4]
ABI	1.11	[1.11,1.40][2]
PWV_{AF} [cm/s]	545	[500,1100][4]
PWV_{CF} [cm/s]	505	[550,1100][5]
PWV_{BD} [cm/s]	564	[800,1400][6]
PWV_{FD} [cm/s]	623	[600,1000][6]

Table 10 Main hemodynamic variables and derived cardiovascular indexes as predicted by the present ADAN model. Reference values are taken preferably for young healthy subjects (< 30 yr) as follows: [1]: Table 1 in Murgo et al (1980), mean of 11 subjects, curves type B and C. [2]: Fowkes et al (2008). [3]: Murgo et al (1980); O'Rourke and Hashimoto (2007). [4]: Liang et al (1998). [5]: Yu et al (2008). [6]: Sugawara et al (2005). HR: heart rate, CO: cardiac output, SBP: systolic blood pressure, DBP: diastolic blood pressure, AI: augmentation index, ABI: ankle-brachial index. PWV: pulse wave velocity. Subscripts (arteries): A: ascending aorta, B: brachial, C: common carotid, F: femoral, D: dorsalis pedis.

6.3 Calibration results

Table 11 presents the PIP solution and the discrepancy in the distribution of flow and in the corresponding resistances when comparing the SAA solution with the PIP solution. This discrepancy is due to the fact that the network introduces a non-negligible impedance which modifies the flow distribution. From the results, it is possible to appreciate that for such a complex network the PIP approach is mandatory to guarantee the correct flow distribution.

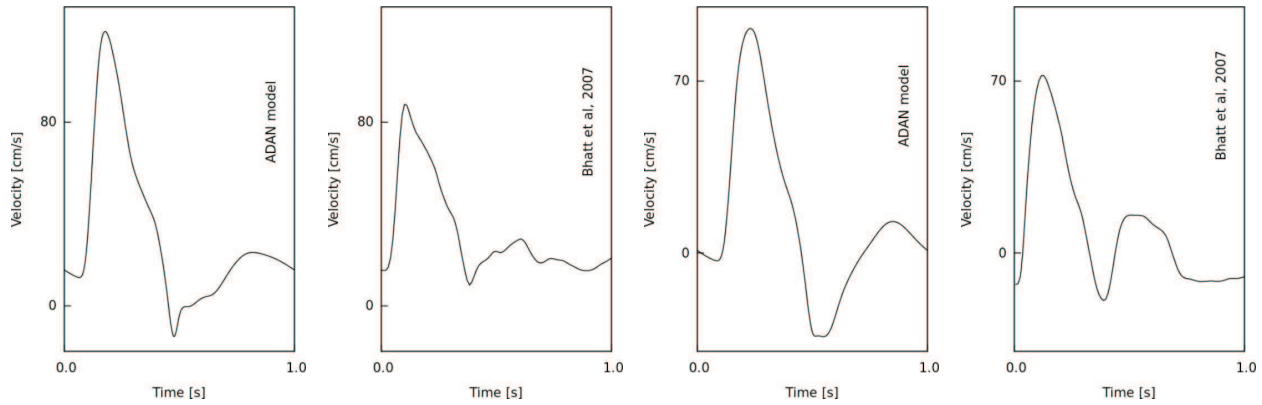
We quantify the discrepancy between the SAA solution and the PIP solution using the following measures

$$e_{R_i} = \frac{|R_i - R_i^{\mathrm{a}}|}{R_i} \qquad i = 1, \ldots, N, \tag{29}$$

$$e_{\bar{\mathcal{Q}}_i} = \frac{|\bar{\mathcal{Q}}_i(\mathbf{R}) - \bar{\mathcal{Q}}_i(\mathbf{R}^{\mathrm{a}})|}{\bar{\mathcal{Q}}_i(\mathbf{R})} \qquad i = 1, \ldots, N, \tag{30}$$

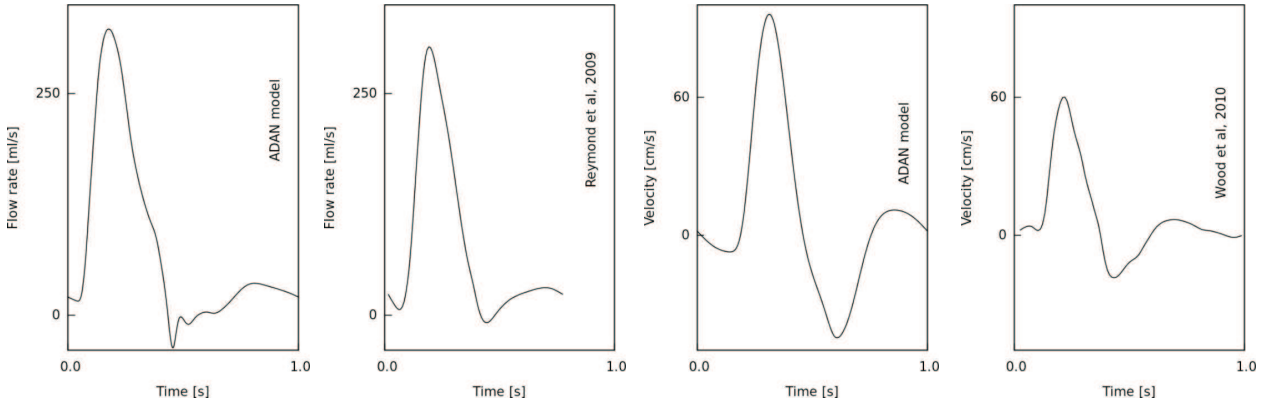
which are reported in Table 11.

Observe that the resistances in the PIP solution are not symmetric and this is due to the lack of strict symmetry of the network. For the vascular territories, maximum discrepancy in the estimation of resistances is over 300%, and in flow is over 30% (see territories C3, right and left). The explanation for the large discrepancy between flow supply in these territories comes from the fact that there are just two arterial vessels supplying



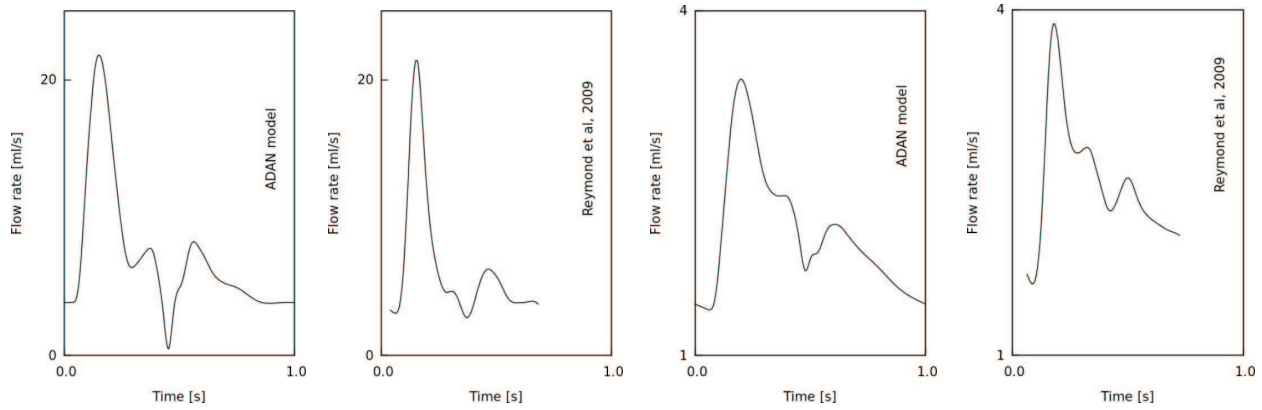
(a) Abdominal Aorta Artery (proximal).

(b) Abdominal Aorta Artery (distal).



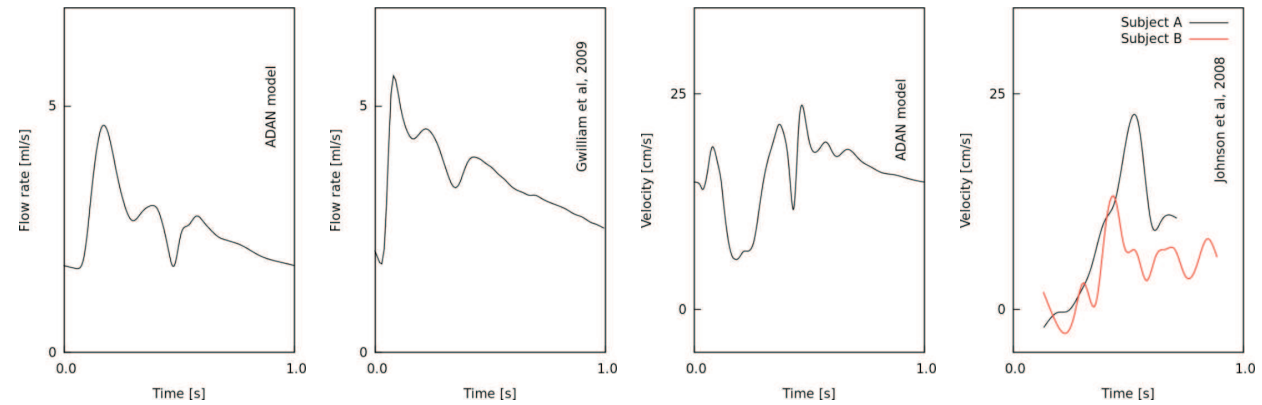
(c) Thoracic Aorta Artery.

(d) Femoral Artery.



(e) Common Carotid Artery.

(f) Middle Cerebral Artery.



(g) Basilar Artery.

(h) Right Coronary Artery.

Fig. 7 Comparison of flow rate and velocity waveforms between the ADAN model and published data.

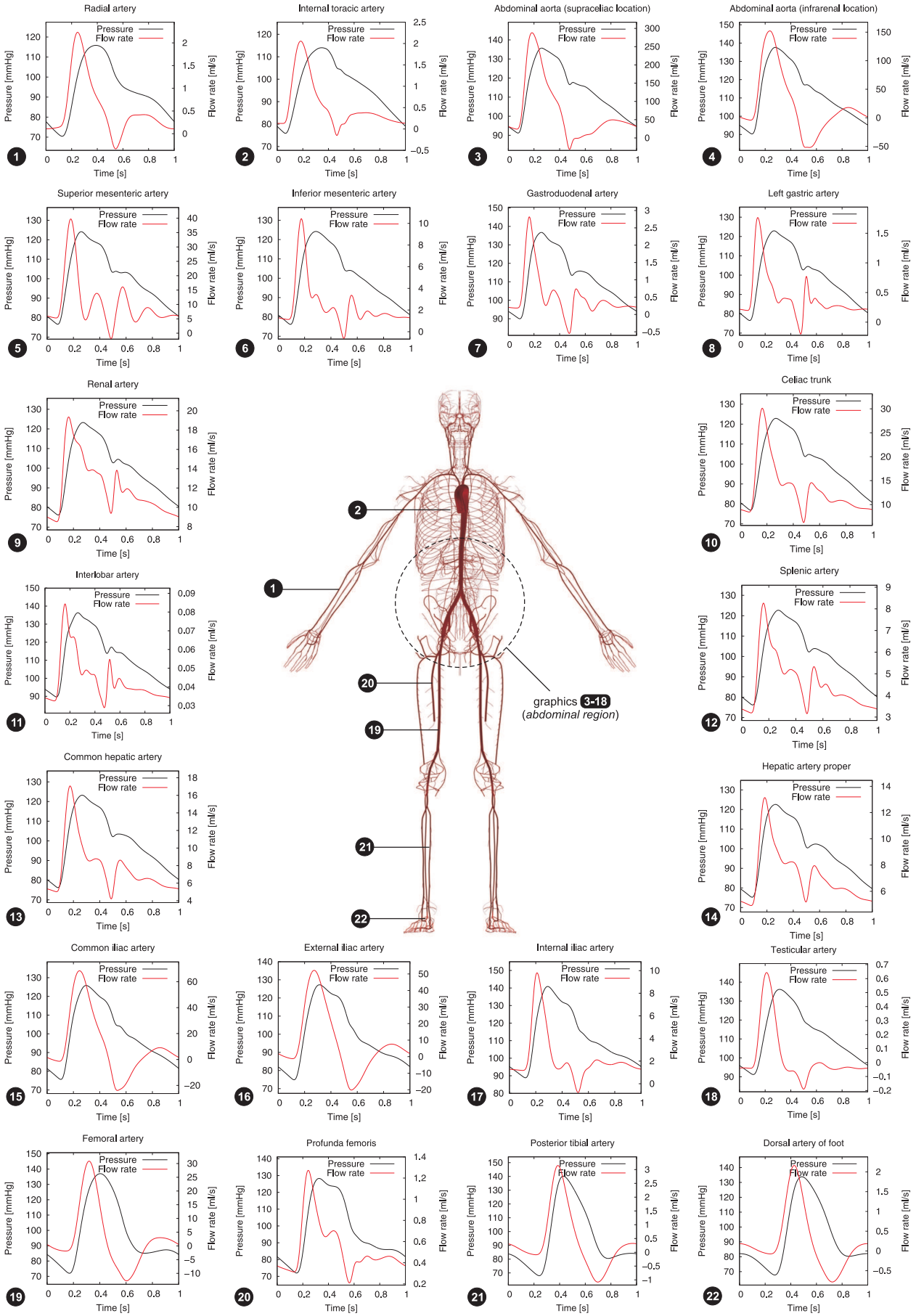


Fig. 8 Pressure (black line) and flow rate (red line) throughout the entire arterial network.

VT	PIP solution		Discrepancy (SAA solution)		VT	PIP solution		Discrepancy (SAA solution)	
	R_i	$\bar{\mathcal{Q}}_i(\mathbf{R})$	e_{R_i} (%)	$e_{\bar{\mathcal{Q}}_i}$ (%)		R_i	$\bar{\mathcal{Q}}_i(\mathbf{R})$	e_{R_i} (%)	$e_{\bar{\mathcal{Q}}_i}$ (%)
A_R1	229768.2	0.494339	17.38	0.55	D_R1	215380.4	0.517807	19.54	1.87
A_R2	246047.1	0.517004	4.81	9.38	D_R2	105605	1.155932	9.22	5.47
A_R3	430813.1	0.278792	11	3.6	D_R3	178074.1	0.615206	21.7	1.93
A_R4	1312793	0.079284	28.09	5.21	D_R4	149843.2	0.762543	16.68	0.17
A_R5	12129307	0.010752	2.23	11.58	D_R5	51444.8	2.10739	22.97	2.93
A_R6	7647416.3	0.017032	2.36	11.45	D_R6	101882.3	0.834707	56.77	14.71
A_R7	699233.2	0.180578	5.59	8.44	D_R7	85384.6	1.36539	14.36	2.02
A_R8	540265.1	0.231001	6.83	7.55	D_R8	650262.4	0.181289	13.09	2.78
A_R9	367530.2	0.338609	7.13	7.51	D_R9	764645.9	0.1677	3.97	9.66
A_R10	1265127	0.094941	11	5.25	D_R10	1781936.7	0.068268	9.6	5.14
A_L1	228570.2	0.494339	17.99	0.12	D_R11	386380.5	0.313931	9.91	5
A_L2	245621.5	0.517004	4.99	9.23	D_R12	1594496.9	0.077718	7.59	6.63
A_L3	430263.2	0.278792	11.14	3.5	D_R13	1191576.1	0.103761	7.83	6.41
A_L4	1310405.3	0.079284	28.32	5.31	D_R14	661366.4	0.17569	14.74	1.6
A_L5	12117916.9	0.010752	2.32	11.51	D_R15	315218.9	0.361474	17.01	0.02
A_L6	7646750.7	0.017032	2.37	11.44	D_R16	517594.4	0.235708	9.28	5.53
A_L7	698115	0.180578	5.76	8.32	D_R17	1444708	0.051716	78.44	20.63
A_L8	539509.7	0.231001	6.98	7.44	D_R18	1419963.6	0.064902	44.67	11.36
A_L9	367041.1	0.338609	7.27	7.41	D_R19	1541370.2	0.075577	14.45	2.12
A_L10	1263915.4	0.094941	11.1	5.18	D_R20	1531598.7	0.076643	13.58	2.72
B1	122381.8	1.036185	5.14	9.02	D_R21	528747.1	0.216566	16.43	0.76
B2	138554.4	0.895781	7.42	7.37	D_L1	213735.5	0.517807	20.46	2.51
B3	930660.6	0.137698	4.04	9.77	D_L2	105608.2	1.155932	9.21	5.48
B4	438397.2	0.286424	6.18	7.89	D_L3	178032.3	0.615206	21.73	1.94
B5	342807.6	0.361631	7.54	6.77	D_L4	149819.9	0.762543	16.7	0.17
B6	1016975.4	0.126043	4.01	9.79	D_L5	51432	2.10739	23.01	2.94
B7	273048.2	0.431229	13.23	2.97	D_L6	101846.2	0.834707	56.83	14.72
B8	166974.7	0.633995	25.94	4.02	D_L7	85365.8	1.36539	14.38	2.02
B9	277793.9	0.427263	12.33	3.58	D_L8	650107	0.181289	13.12	2.77
B10	177971	0.707759	5.84	8.39	D_L9	764499.5	0.1677	3.99	9.65
B11	2005919.4	0.062808	5.82	8.49	D_L10	1781532.4	0.068268	9.62	5.13
B12	457729.4	0.243295	19.72	0.89	D_L11	386290.9	0.313931	9.94	4.99
B13	198475.8	0.462976	45.09	10.88	D_L12	1594139.6	0.077718	7.61	6.62
B14	1224867.1	0.105446	3.22	10.79	D_L13	1191309.2	0.103761	7.86	6.41
B15	153516.6	0.82756	4.94	9.31	D_L14	661163.6	0.17569	14.77	1.59
B16	767672.4	0.161973	7.22	7.05	D_L15	315110.6	0.361474	17.05	0
B17	788123.6	0.158181	6.94	7.29	D_L16	517436.8	0.235708	9.31	5.52
B18	176320.9	0.696443	8.57	6.33	D_L17	1443691.5	0.051716	78.57	20.64
B19	609079.7	0.187986	16.44	0.91	D_L18	1419228.5	0.064902	44.74	11.38
B20	3385229.7	0.037262	5.69	8.55	D_L19	1540765.1	0.075577	14.49	2.1
B21	2497603.3	0.050298	6.13	8.18	D_L20	1531008.5	0.076643	13.62	2.7
B22	5213365.6	0.024689	3.58	10.48	D_L21	528535.3	0.216566	16.48	0.73
C_R1	328815.7	0.311233	30.28	6.26	O1	22991.4	4.484667	4.23	12.96
C_R2	516288.6	0.229018	12.76	2.91	O2	6566.9	13.454	50.9	12.71
C_R3	63921.2	0.472101	341.8	30.67	O3	7177493.5	0.016667	11.45	5.37
C_R4	321360.4	0.336038	23.46	2.95	O4	7171951.1	0.016667	11.54	5.31
C_R5	331330.9	0.364884	10.28	4.73	O5	7903671206	0.000016	3.05	10.67
C_R6	397455.8	0.205908	62.91	16.07	O6	7892472837	0.000016	3.2	10.56
C_R7	407429.6	0.257629	27.02	4.96	O7	1267654216	0.000103	1.74	11.86
C_R8	954673.6	0.119374	16.99	0.27	O8	372428.5	0.35	2.28	11.59
C_R9	220775.5	0.490828	23.03	3.21	O9	93354157.7	0.0014	1.99	11.75
C_R10	248231.6	0.4647	15.58	1.09	O10	74222.8	1.68175	6.81	7.69
C_R11	740478.7	0.158946	13.28	2.67	O11	10604602	0.011	14.29	3.38
C_R12	2702565	0.040204	22.7	3.1	O12	15161	7.287583	20.67	1.64
C_R13	709392.6	0.145734	28.96	6.27	O13	25698898.4	0.005	3.76	10.67
C_R14	1914169.6	0.062583	11.29	4.09	O14	11821.4	10.651083	5.89	9.38
C_R15	2619998.3	0.045177	12.64	3.02	O15	11980.8	10.651083	4.48	9.79
C_R16	518550.1	0.219725	17.01	0.06	O16	737242	0.168175	7.53	7.18
C_L1	328242.4	0.311233	30.5	6.37	O17	747963.1	0.168175	5.99	8.47
C_L2	515608.4	0.229018	12.91	2.81	O18	114281.7	1.121167	4.05	10.07
C_L3	63409.8	0.472101	345.36	30.75	O19	98912	1.121167	20.22	1.38
C_L4	320843.9	0.336038	23.66	3.06	O20	38637	3.3635	2.59	11.37
C_L5	330874.6	0.364884	10.43	4.61	O21	6239.4	11.211667	90.59	20.42
C_L6	396515.3	0.205908	63.29	16.18	O22	32888.2	3.643792	11.25	4.26
C_L7	406760.8	0.257629	27.22	5.07	O23	1929787.3	0.06727	2.7	11.09
C_L8	953199	0.119374	17.17	0.14	O24	92364.1	1.041997	38.53	9.17
C_L9	220750.6	0.490828	23.05	3.23	O25	3873441.8	0.033542	2.62	11.4
C_L10	248031.7	0.4647	15.67	1.02	O26	3873731.9	0.033542	2.61	11.41
C_L11	739701.8	0.158946	13.4	2.58	O27	149815.2	0.840875	5.83	8.32
C_L12	2703359.5	0.040204	22.67	3.09	O28	100381	1.235171	7.53	7.2
C_L13	709635.8	0.145734	28.92	6.26					
C_L14	1915085.1	0.062583	11.24	4.11					
C_L15	2620413.6	0.045177	12.62	3.02					
C_L16	518592.2	0.219725	17	0.05					

Table 11 Resistances and actual flow in vascular territories and specific organs for PIP solution and the discrepancy encountered in the results when using the SAA solution. VT: vascular territory, SAA: simple analytical approach, PIP: parameter identification problem. Flows are in cm^3/s and resistances are in $dyn\,s/cm^5$.

those territories, which makes the SAA a very inexact method to compute the resistance such that the blood flow is the required one. This is an example of the deficiency of the SAA method which is overcome by the proposed PIP strategy. In the specific organs, the discrepancy is over 90% and 20% (see specific organ O21, the small intestine), respectively.

In average, for vascular territories discrepancy in the estimation of resistances is 22.5%, and in flow is 6.1%. Analogously, for the specific organs the averaged discrepancy is 12.4% and 9.2%, respectively. Evidently, with the algorithm proposed in the present work which delivers the PIP solution this discrepancy is reduced below the convergence tolerance of the Newton method. With this data at hand it is possible to analyze whether the PIP solution of equation (16) is required or not, according to the level of error which is considered acceptable in the analysis. For example, the pressure at the aortic root is not sensitive to changing from the PIP solution to SAA solution, while the flow through smaller arteries could vary significantly, as seen in Table 11.

6.4 Sensitivity analysis

In this section we present a basic sensitivity analysis to show the effect of the main parameters of the model in model predictions. Firstly, we analyze the sensitivity with respect to the factors θ_R and θ_C which define the ratio between Windkessel resistances and the terminal compliance. Secondly, we analyze the effect of modifying the arterial wall parameters E_e, E_c, ε_0 and ϵ_r by applying, correspondingly, multiplicative factors θ_e, θ_c, θ_0 and θ_r.

Sensitivities are presented at three locations: (a) a central artery, the abdominal aorta artery (AAA), (b) a secondary artery, the internal carotid artery (ICA), and (c) a more distal artery, the posterior tibial artery (PTA). These locations are representative locations of the sensitivities observed throughout the entire network.

6.4.1 Sensitivity to terminal parameters

We consider the ratio $\theta_R \in \{0.1, 0.2, 0.5, 0.8, \text{matched}\}$, where $\theta_R = 0.2$ is the control value chosen for the simulations in Section 6.2. The case $\theta_R = $ matched stands for R_A equal to the characteristic impedance of the arriving arterial segment.

Figure 9 shows the pressure and flow waveforms for the different cases considered. From the results it is clear that the pressure pulse is not sensitive to θ_R, while the flow rate displays a small sensitivity in the AAA and in the PTA. The sensitivity is slightly higher in the ICA because of the larger amount of blood required by the brain. At the ICA differences are concentrated during systolic flow decay, and during diastolic flow uprise. This is an evidence of those periods in which parameters of the Windkessel elements can affect waveforms (aside from the total resistance which evidently affects the mean blood flow rate carried by arteries). In turn, invariant features are due to the particular form of the ejection heart boundary condition, i.e. the systolic upstroke, the dicrotic notch and the final diastolic decay. Those features are not sensitive to θ_R.

Noting that the ratio θ_R can not go to much beyond the studied limits, we conclude that the sensitivity of model predictions with respect to θ_R is negligible.

Now, we analyze the effect of the peripheral compliance by setting $\theta_C \in \{0, 0.05, 0.1\}$, noting that $\theta_C = 0.1$ is the control case for results in Section 6.2.

Figure 10 presents the pressure and flow waveforms at the considered locations. In this case, similar conclusions hold for the sensitivity with respect to θ_C. Pressure is not sensitive to θ_C, while the peak flow rate during systole slightly varies, with more sensitivity in the ICA for the same reason exposed above. The most sensitive periods during the heart beat are, as before, during systolic decay and during diastolic uprise, while invariant features such as upstroke, dicrotic notch and diastolic decay remain the same.

From this case we conclude that, given that we have considered and arbitrary but reasonable upper bound for the peripheral compliance, the sensitivity of model predictions is small with respect to θ_C.

The results derived in the present study are not surprising. Due to the degree of completeness of the network the setting of outflow boundary conditions results a far more easy task. In fact, this indicates that a hypothesis of purely resistive terminal elements would be fully justified in this highly detailed network.

6.4.2 Sensitivity to wall parameters

The factors that affect the baseline wall parameters to perform the sensitivity analysis are taken to be $\theta_x \in \{0.7, 1.0, 1.3\}$, $x \in \{e, c, 0, r\}$ (e.g., we have $E_e^* = \theta_e E_e$, $\varepsilon_0^* = \theta_0 \varepsilon_0$, and so on). Observe that the unitary value corresponds to the control case. We perform canonical perturbations in the universe of parameters, i.e., when one parameter is modified the others remain invariant.

Figure 11 presents the sensitivity with respect to θ_e. This parameter is crucial to understand the interactions of waves that ultimately define the pressure and flow waveforms. The sensitivity is large at central and peripheral locations. Increase of θ_e results in stiffenning,

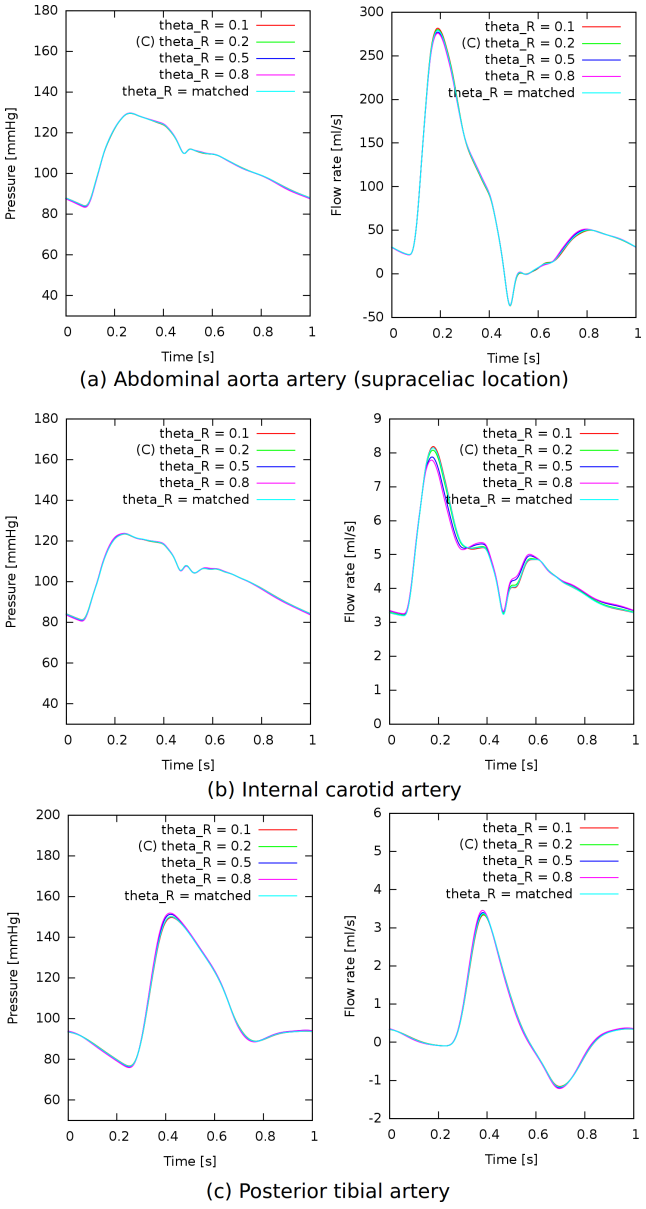


Fig. 9 Sensitivity with respect to the ratio between Windkessel resistances ruled by θ_R. Pressure (left) and flow rate (right) are shown for three distant locations. Control case is denoted by (C).

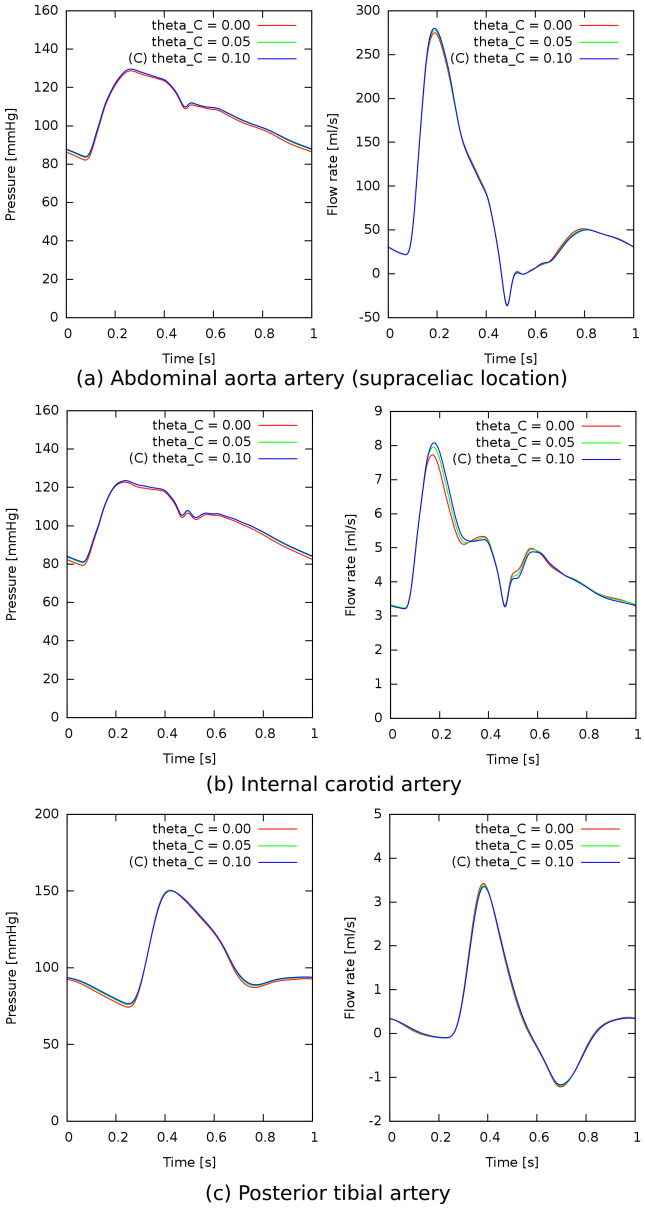


Fig. 10 Sensitivity with respect to the terminal compliance ruled by θ_C. Pressure (left) and flow rate (right) are shown for three distant locations. Control case is denoted by (C).

increasing the pressure upstroke, and anticipating the pressure pulse due to the increase of the pulse wave velocity. Decreasing θ_e results in a smaller reflection right after the dicrotic notch (more visible at the ICA and PTA), and changes the characteristic time that determines the diastolic decay. The upstroke in flow waveform remains invariant for the AAA and ICA, but is significantly affected at the PTA because of its more distal location. Systolic flow is reduced for larger E_e, and the anticipated diastolic flow uprise for larger θ_e visible at the PTA is manifested as an anticipated reflection during diastole at the AAA.

Figure 12 presents the sensitivity with respect to the stiffness of collagen fibers, that is θ_c. For the present arterial wall model the sensitivity of the predictions is small with respect to this parameter. However, even being small sensitivities, it is remarkable the fact that for larger θ_c, the systolic flow increases (unlike for larger θ_e). As well, by stiffenning the collagen (larger θ_c), the foot of the pressure pulse does not change as significantly as when stiffenning the elastin (larger θ_e). This tells us that the collagen fibers are indeed being recruited during systolic upstroke (see pressure at PTA). This will be more evident in the forthcoming analysis.

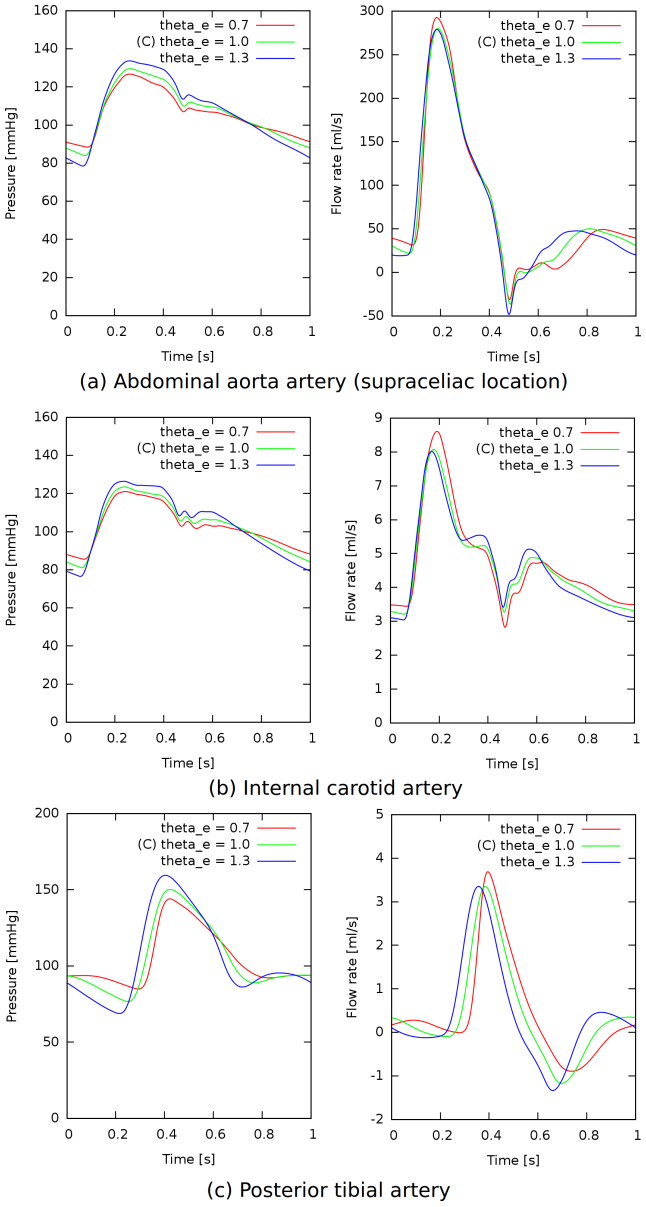


Fig. 11 Sensitivity with respect to the stiffness of the linear elastic component (elastin stiffness) ruled by θ_e. Pressure (left) and flow rate (right) are shown for three distant locations.

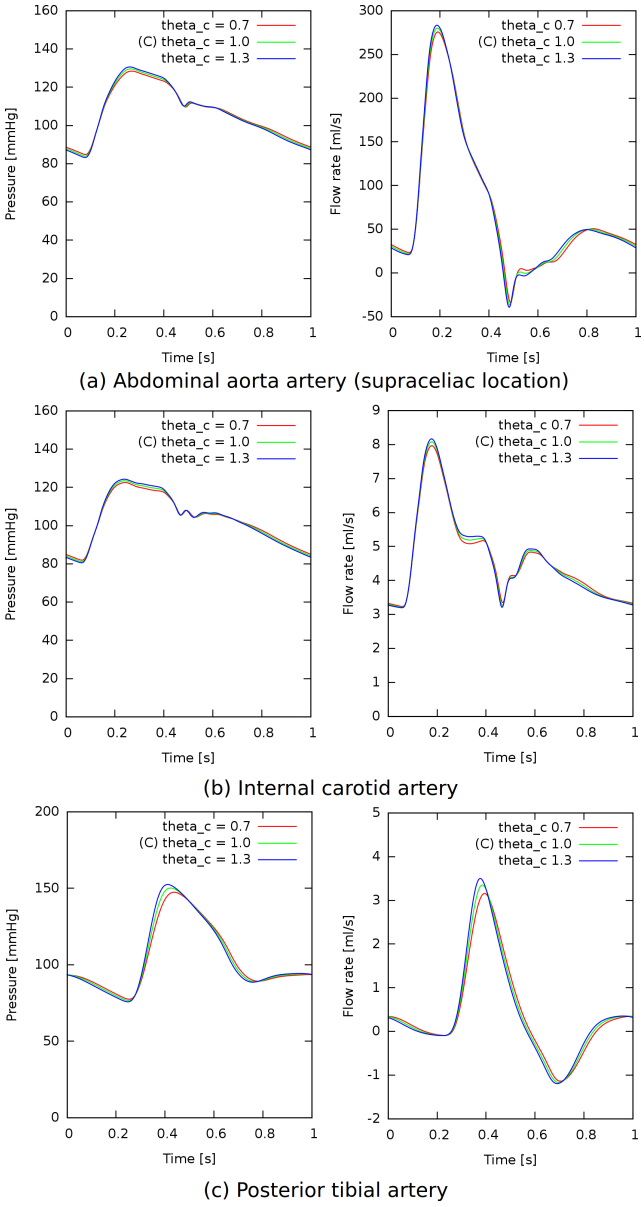


Fig. 12 Sensitivity with respect to the stiffness of the non-linear elastic component (collagen stiffness) ruled by θ_c. Pressure (left) and flow rate (right) are shown for three distant locations.

Figure 13 and Figure 14 present the sensitivity with respect to the parameters controling the baseline strain (θ_0) and to the strain deviation (θ_0), which determine the central position and the width of fiber recruitment distribution, respectively. Sensitivities are high in this case, posing the relevance of a non-linear model to account for collagen fiber. Remarkably, just by introducing the collagen contribution into our model it is possible to predict pressure pulse and flow waveforms changes like those observed in disease. Smaller values of θ_0 (as well as larger values of θ_r) imply more fibers recruited for a fixed level of applied strain. As stated by Burton (1954), these results support the need for complex arterial wall models to account for non-linear effects for modeling disease conditions.

According to the calibration of our model, the sensitivity is higher when reducing θ_0 or increasing θ_r. At all locations we can observe changes in the amplitude of the pressure pulse, changes in the flow upstroke as well as during systolic decay. Reflections around the dicrotic notch are also affected. Observe that, unlike the sensitivity with respect to θ_e, in this case, a stiffer network (reduced θ_0 or increased θ_r) amounts to an increase in

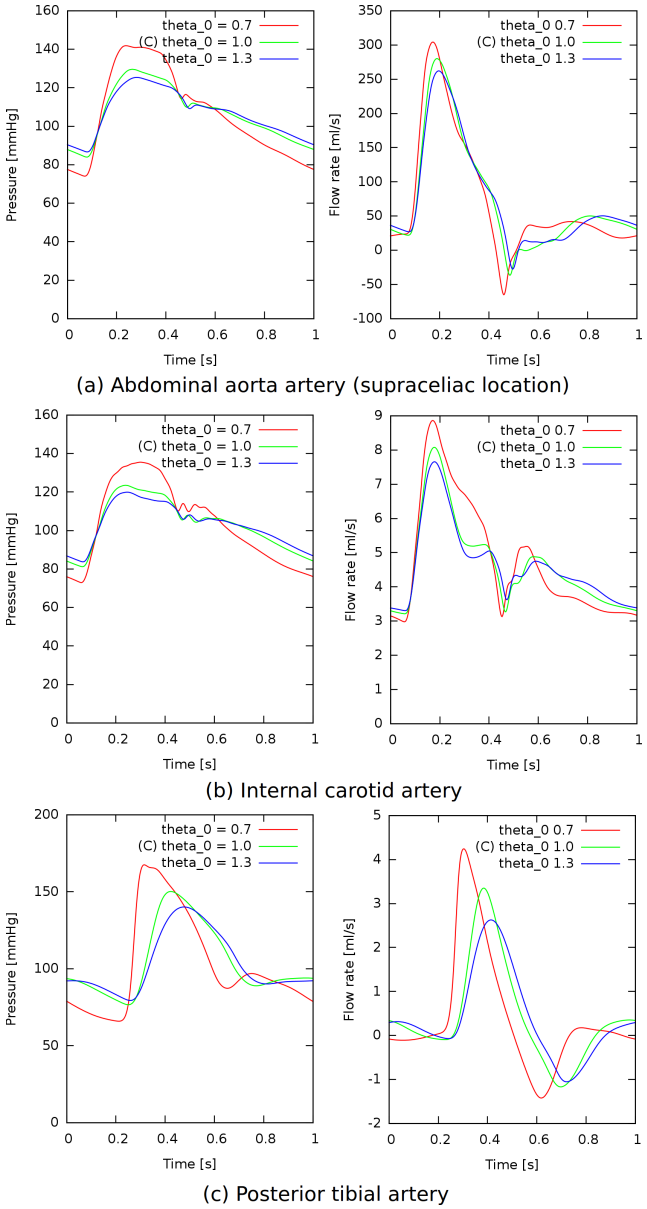


Fig. 13 Sensitivity with respect to the parameter that defines the baseline strain for collagen fiber recruitment ruled by θ_0. Pressure (left) and flow rate (right) are shown for three distant locations.

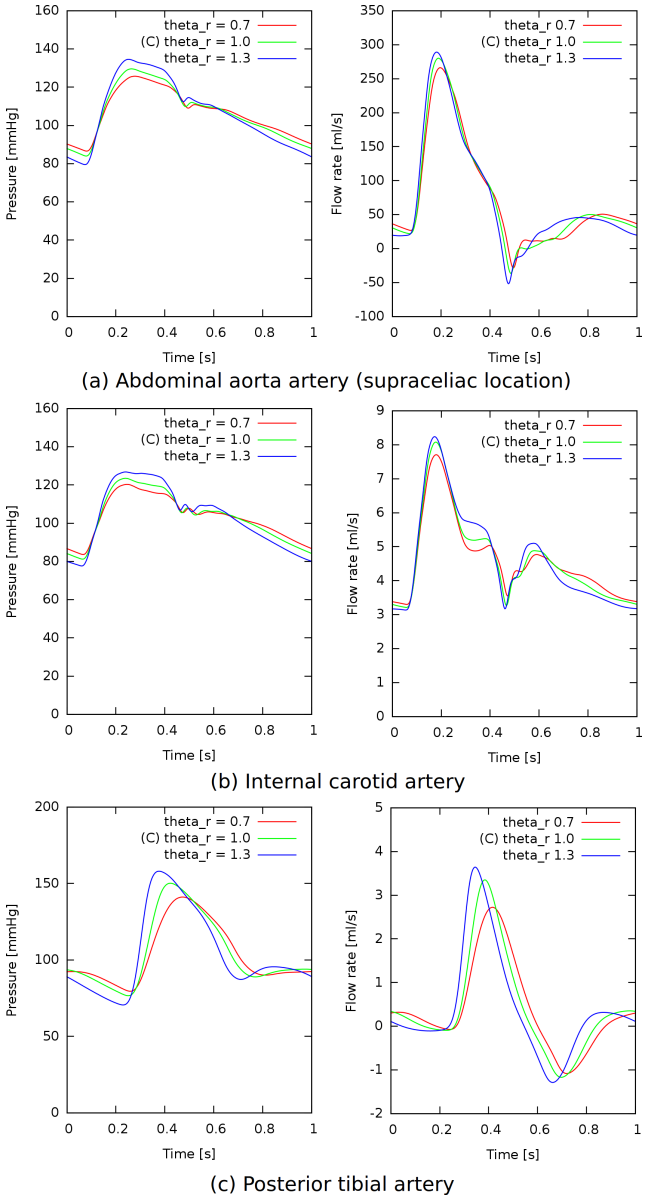


Fig. 14 Sensitivity with respect to the parameter that defines the standard deviation in strain for collagen fiber recruitment distribution ruled by θ_r. Pressure (left) and flow rate (right) are shown for three distant locations.

the systolic flow and a decrease in the minimum flow reached at the beginning of diastole.

The results presented in this section justify the use of a composite arterial wall model including elastin and collagen as different components. The complex interplay between those two components is responsible for the conformation of arterial pressure pulse and flow waveforms throughout the entire tree. In this sense, the accurate determination of these parameters poses a challenge from the modeling point of view (inverse parameter estimation) and from the experimental point of view.

6.5 Extensions and future research

Other cardiovascular scenarios were tested by modifying model parameters and keeping the same blood flow distribution. In all tested cases, the solution of equation (16) using the Newton method was successfully achieved, demonstrating the ability of the proposed strategy in estimating terminal parameters.

At this point it is important to recall the goals of the present contribution. The first goal has been to develop criteria to determine the blood distribution in a normal systemic condition. With this data at hand, the

second goal has been to formulate a Newton-based algorithm to compute a set of terminal resistances such that the given blood flow distribution is met. In all tested cases we have not changed the criteria to distribute flow. These criteria encapsulate the behavior and state of the peripheral vasculature, while the numerical strategy based on the Newton method is just responsible for computing the corresponding terminal parameters that are such that the blood flow is supplied as given by these criteria (or by any other which could have also been defined). In cases of altered systemic conditions, such as hypertension, diabetes, renal disease, liver dysfunction, peripheral vascular disease, etc., or also in case of other physiological conditions such as exercise, after meals, fasting, rem sleep, sleep-wake action, etc., the criteria to distribute blood flow should be revised, providing proper modifications according to the analyzed condition. In fact, altered arterial functioning is not only a property of large and medium-sized vessels. When altered conditions are observed in the larger arteries, abnormal conditions are taking place in the peripheral beds. This is what poses the challenge of revisiting/checking the hypothesis behind the criteria to distribute blood in cases of disease in order to be able to predict flow and pressure waveforms.

A word on uncertainty analysis is in order. Assuming that blood flow distribution criteria and that geometrical parameters (vessel radius and wall thickness) are correctly determined, the dominant uncertainties of the model are the inflow boundary condition and the parameters of the arterial wall model, more specifically E_e, ε_0 and ϵ_r from our model (see Section 6.4). The former is important in the determination of main features of pressure pulse and upstroke in the central arteries, such as systolic upstroke and location of dicrotic notch, while the latter mostly affects the pressure pulse formation as well as small features of flow waveforms such as systolic decay and diastolic flow uprise, among others. It is worth mentioning that the present highly detailed model is also a computational laboratory to analyze complex systemic interactions which could lead to uncertainty quantification as done by Chen et al (2013). In such work, it is claimed that a more detailed description of the arterial network could improve the prediction of pressure waveforms. Several conclusions from that work would remain valid for the present model. However, conclusions drawn of the relation of pressure/flow and model parameters at peripheral locations in which collateral circuits are present may rely on more intricate explanations, and eventually change. In any case, such uncertainty analysis on top of the present model would constitute an important advance in the field.

7 Final remarks

In this work we have presented the criteria to determine the blood flow distribution in an anatomically detailed model of the arterial network, which is mandatory towards performing numerical simulations of arterial blood flow within such a complex model. This comprised criteria to define blood flow to specific organs as well as to determine regional blood supply to vascular territories. In the later case, the underlying hypotheses and their link to physiological and anatomical concepts have been clearly exposed and fully justified. Furthermore, a numerical methodology was presented to guarantee that such flow distribution is satisfied in numerical simulations performed over this detailed model of the arterial vasculature. Such approach relied on the formulation of a parameter identification problem which, when solved with the Newton method, proved to be successful in computing the parameters of terminal models that are involved in the determination of the blood flow distribution for such model.

It is worthwhile to point out that the focus of the present contribution is on the definition of criteria to define flow distribution and on the numerical strategy to perform the calibration of peripheral model parameters. A detailed account of the fidelity of the arterial topology, and calibration of arterial vessels parameters is a matter of another contribution.

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